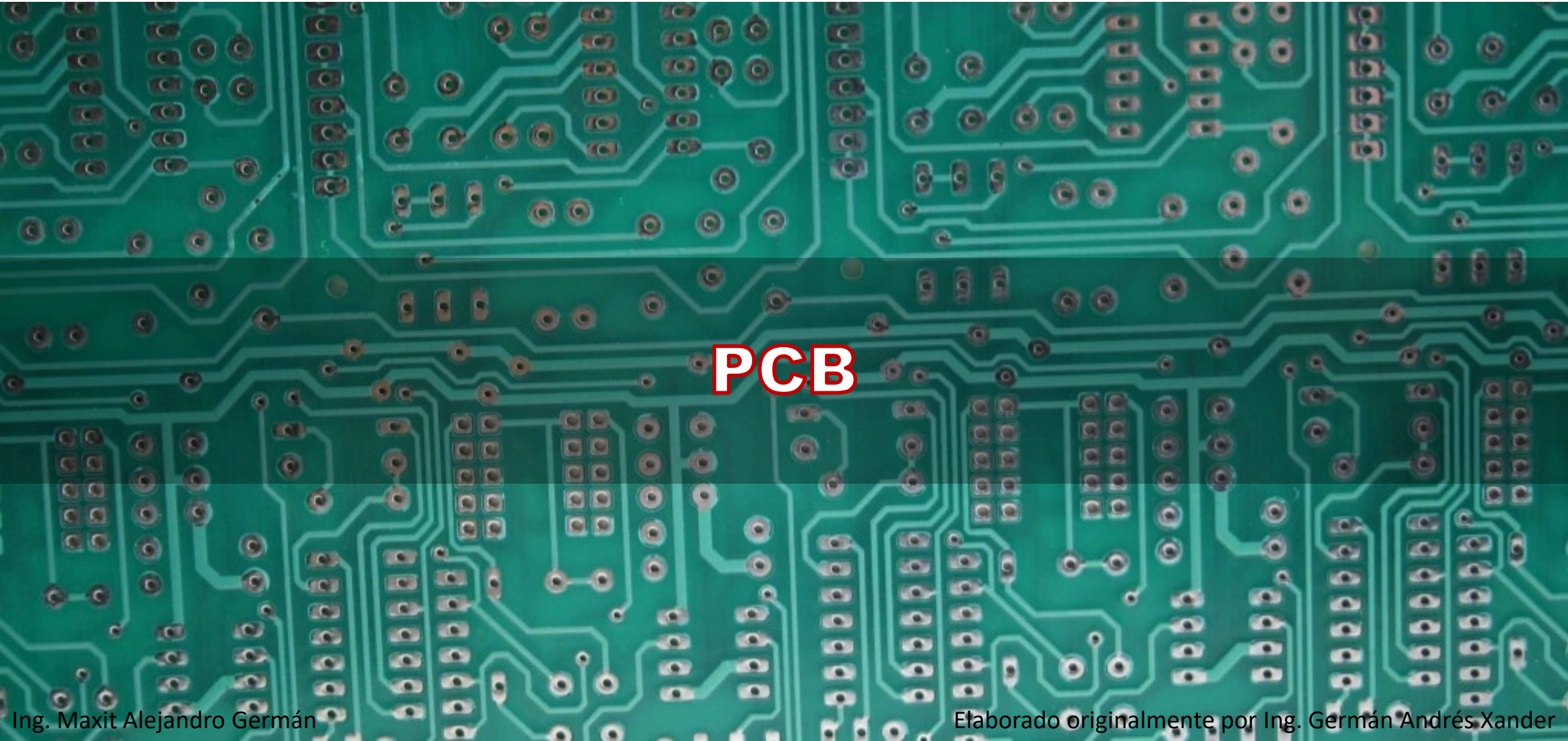
UNIVERSIDAD NACIONAL DE MISIONES

INTRODUCCIÓN A LA TECNOLOGÍA MECATRÓNICA [IM205]

Facultad de Ingeniería OBERA



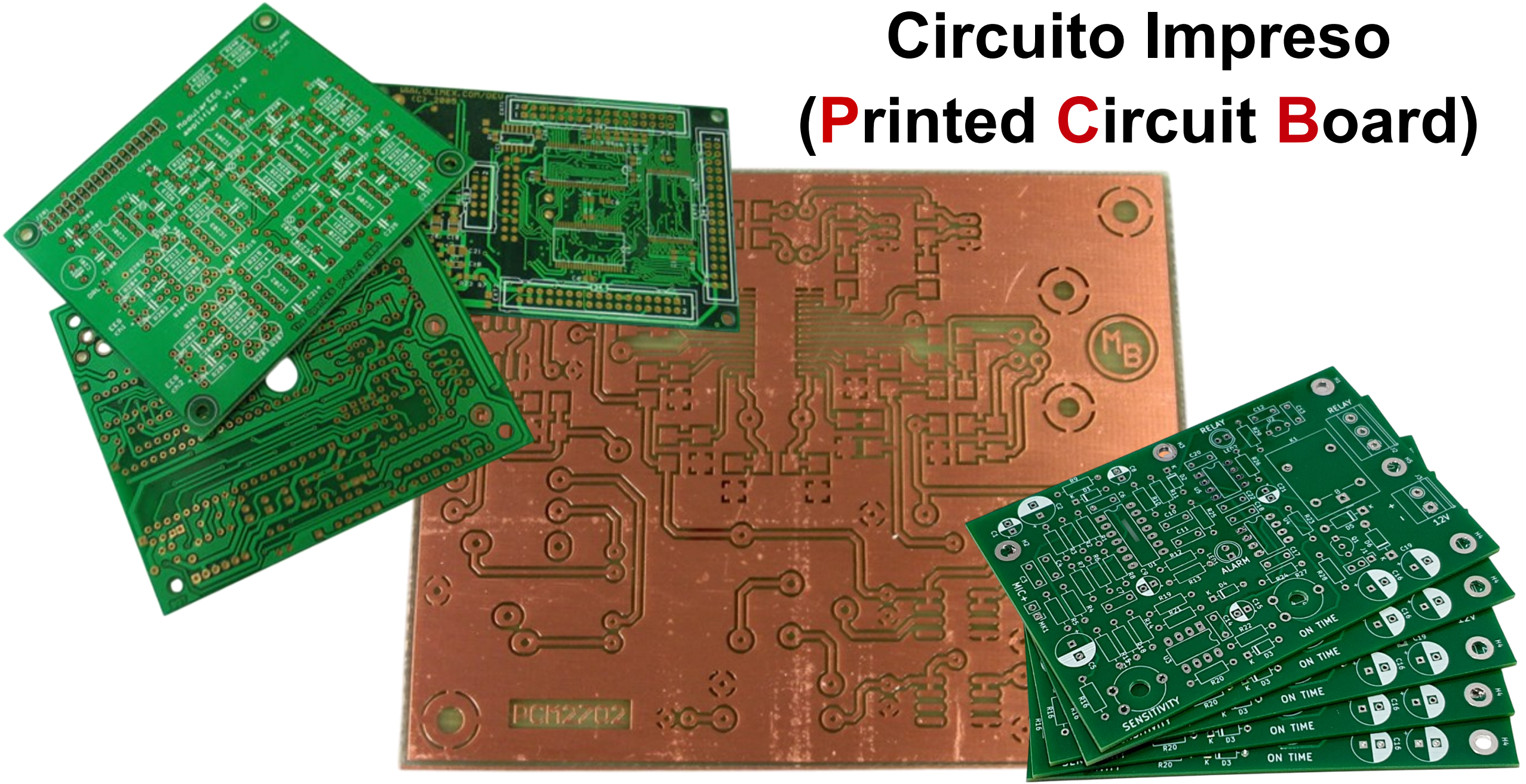
PCB

Ing. Maxit Alejandro Germán

Elaborado originalmente por Ing. Germán Andrés Xander

PCB

Circuito Impreso
(Printed Circuit Board)



ESTRUCTURA SIMPLE FAZ

ESTRUCTURA SIMPLE FAZ



Cobre

Sustrato

- Fenólicos
- Fibra de Vidrio

ESTRUCTURA SIMPLE FAZ



Cobre

Sustrato





ESTRUCTURA DOBLE FAZ



ESTRUCTURA DOBLE FAZ



Espesor cobre

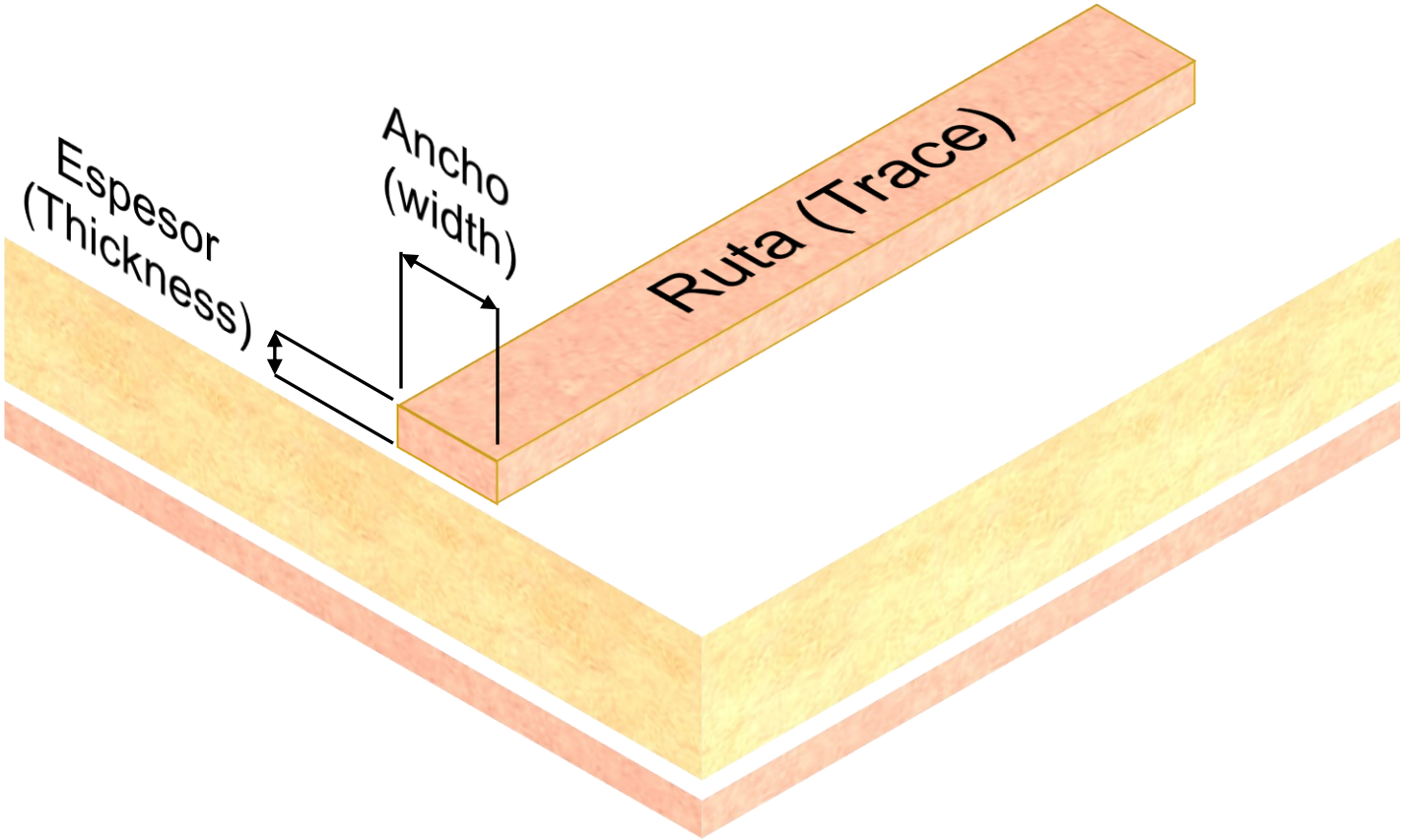
oz/ft²	µm	in	mil
0,5	17,5	0,0007	0,7
0,75	25,5	0,0010	1,0
1	35	0,0014	1,4
2	71	0,0028	2,8
3	105	0,0042	4,2

ESTRUCTURA DOBLE FAZ

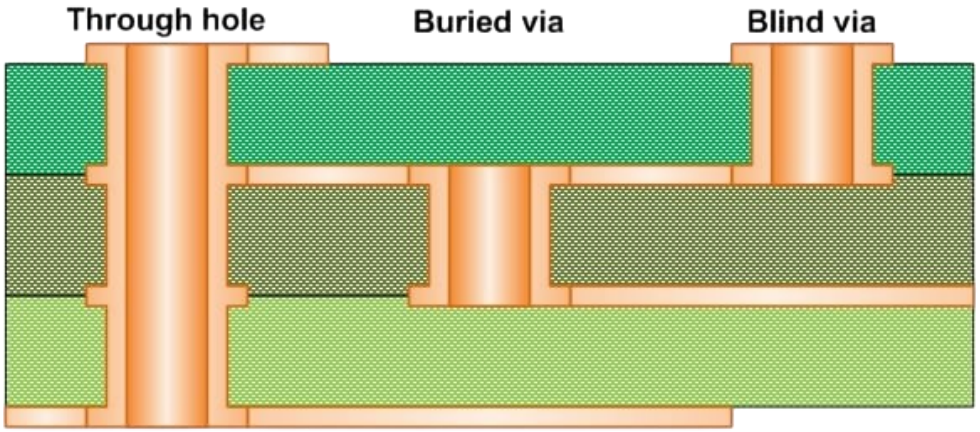


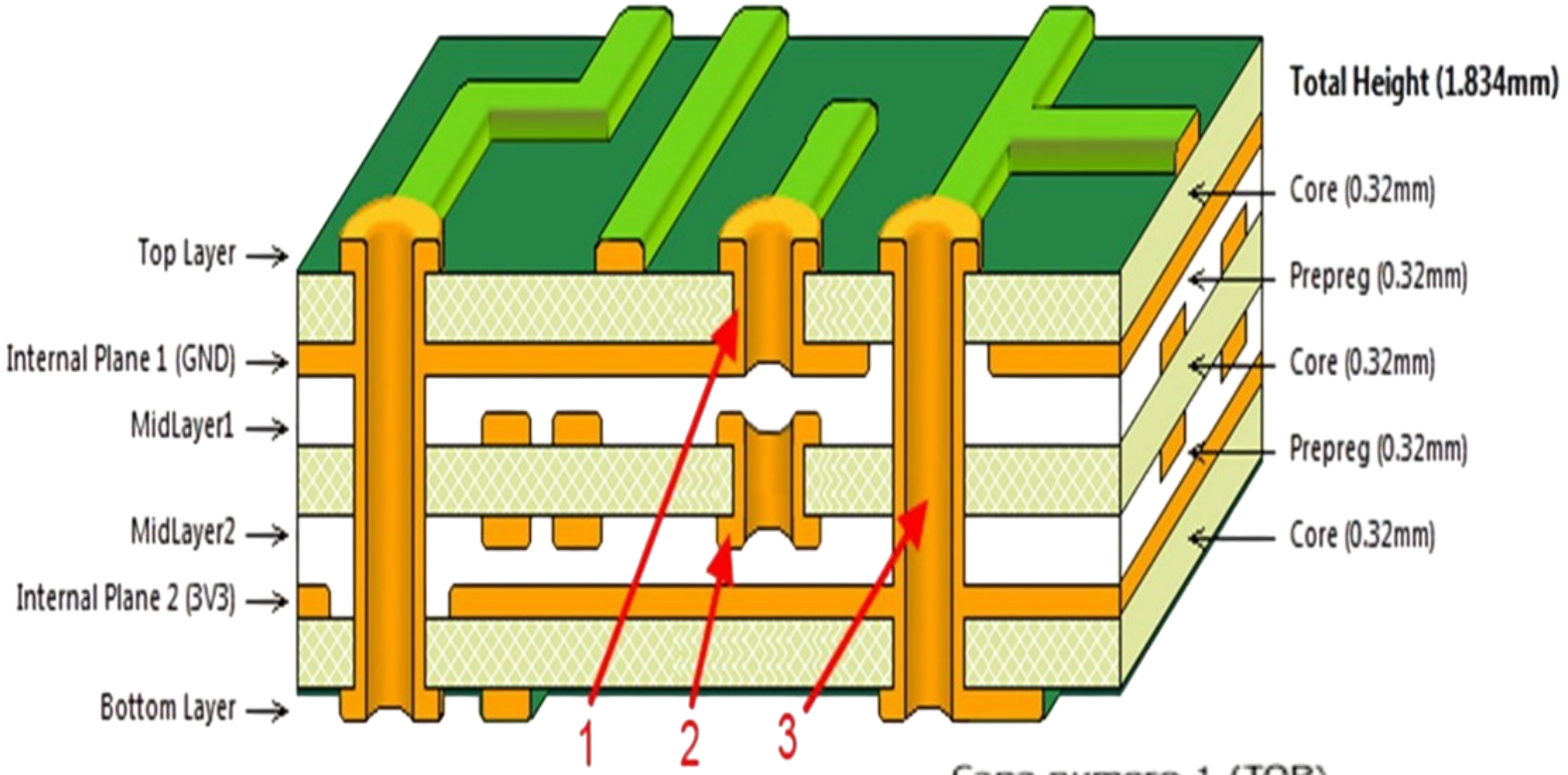
Espesor cobre

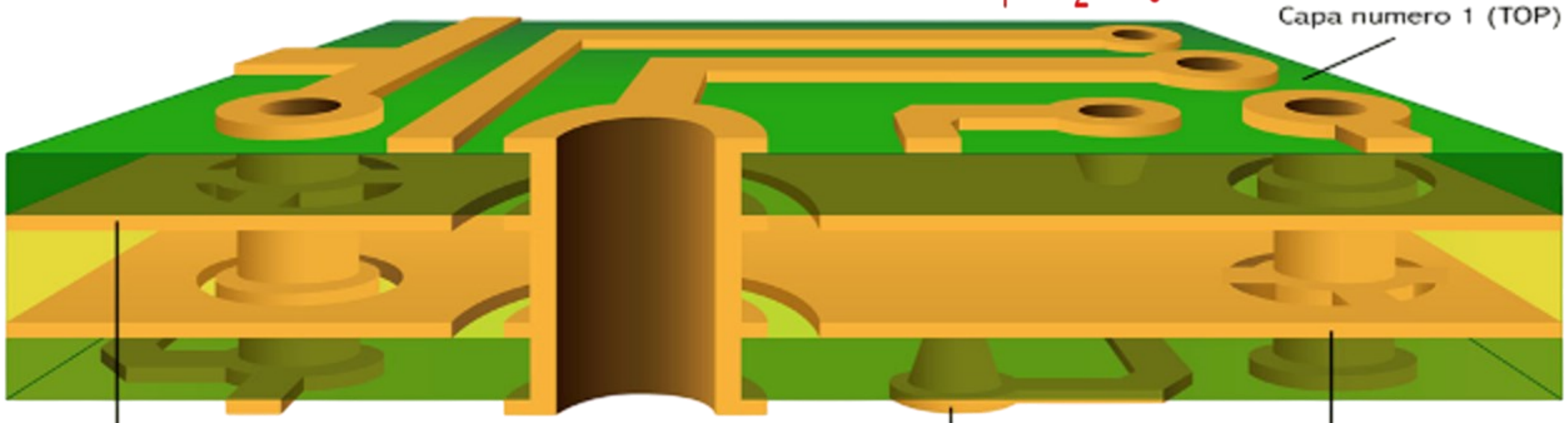
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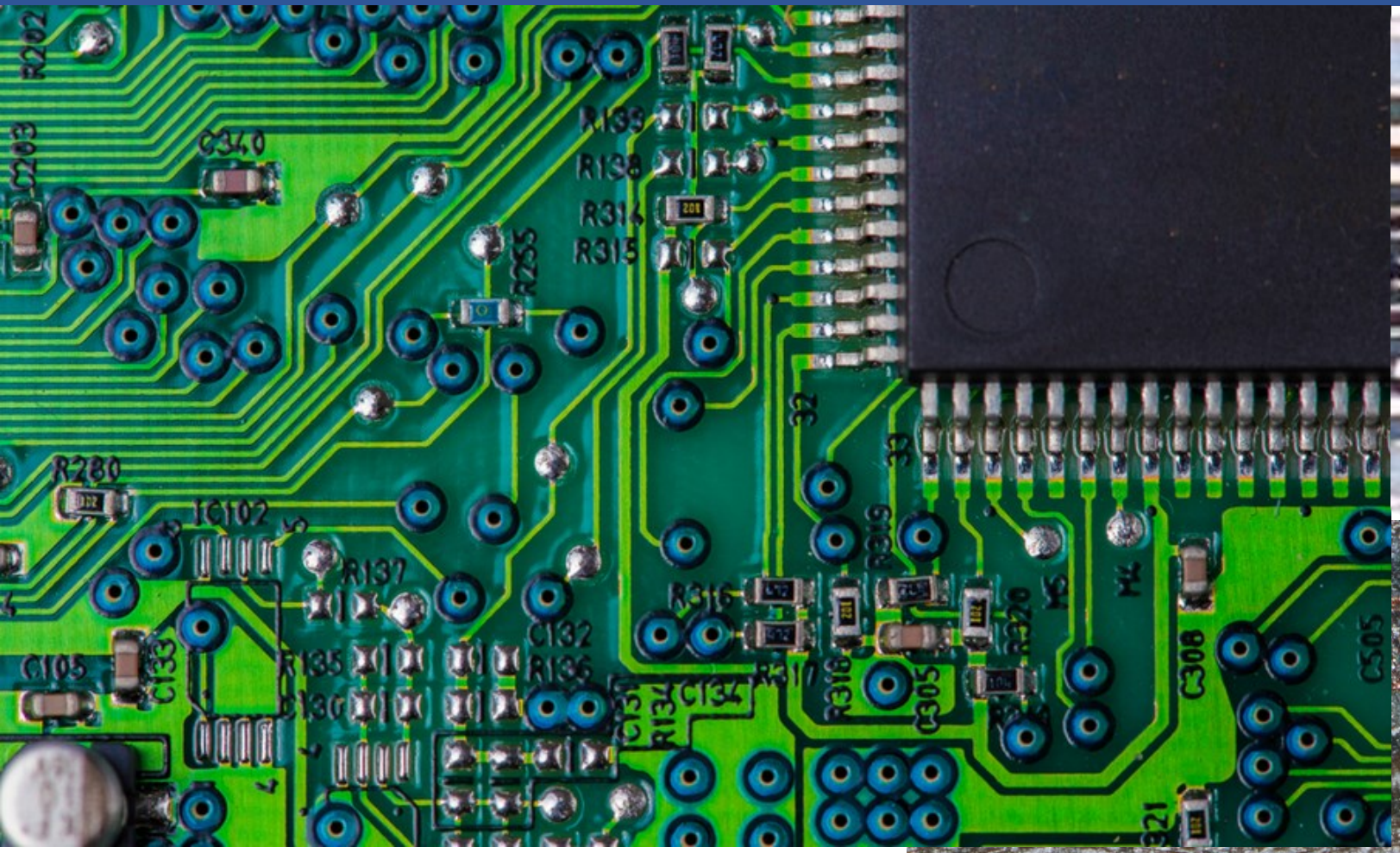
MULTICAPA

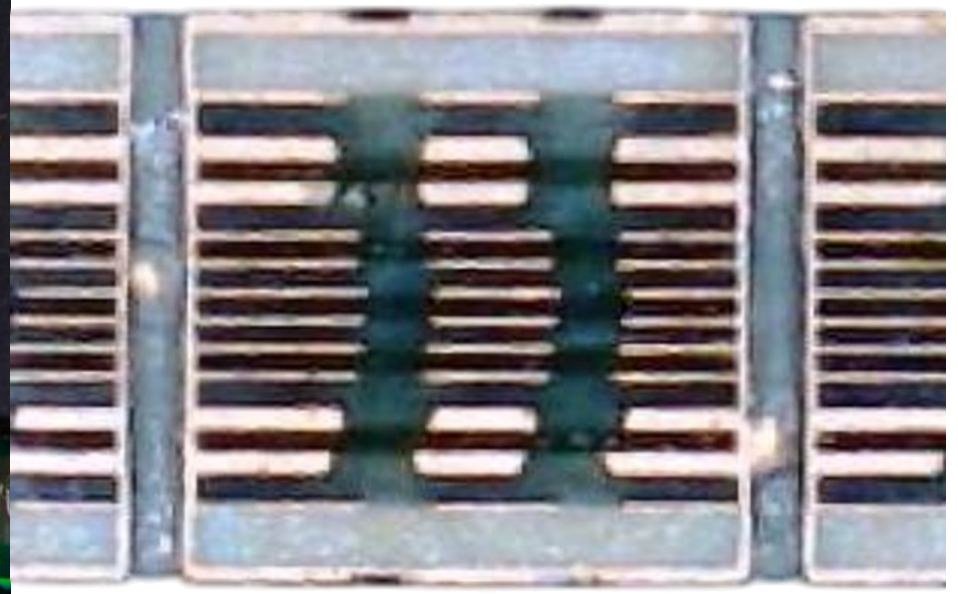

Through hole
Buried via
Blind via

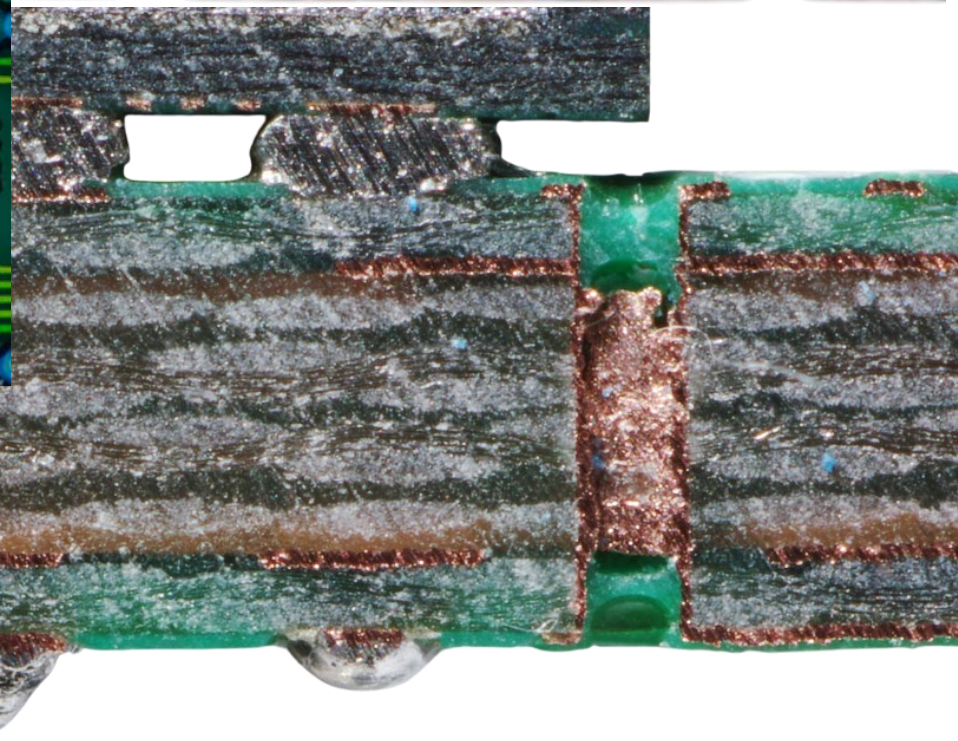

Total Height (1.834mm)
Top Layer
Internal Plane 1 (GND)
MidLayer1
MidLayer2
Internal Plane 2 (3V3)
Bottom Layer
Core (0.32mm)
Prepreg (0.32mm)
Core (0.32mm)
Prepreg (0.32mm)
Core (0.32mm)
1
2
3


Capa numero 1 (TOP)

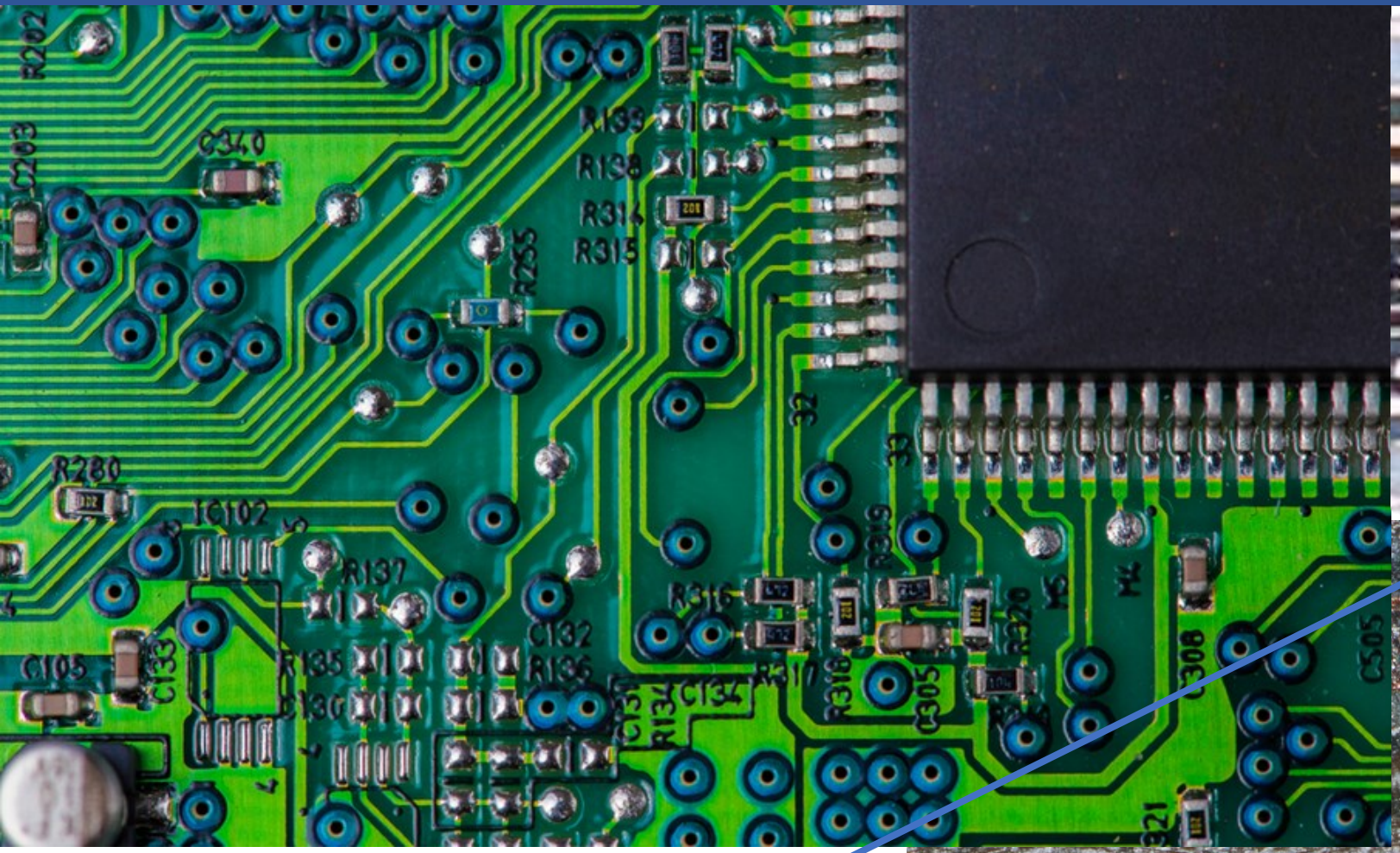
MULTICAPA

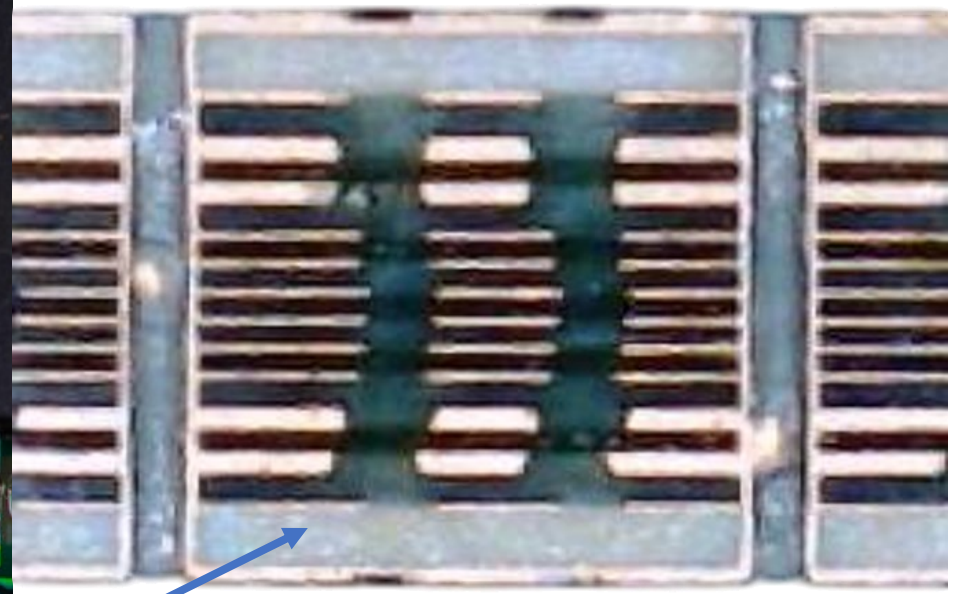

R202
C203
C340
R133
R138
R314
R315
R255
R280
IC102
R137
C132
R136
R135
C133
C130
C105
C131
R134
C134
R316
R317
R318
R319
C305
R320
C308
C505
R321

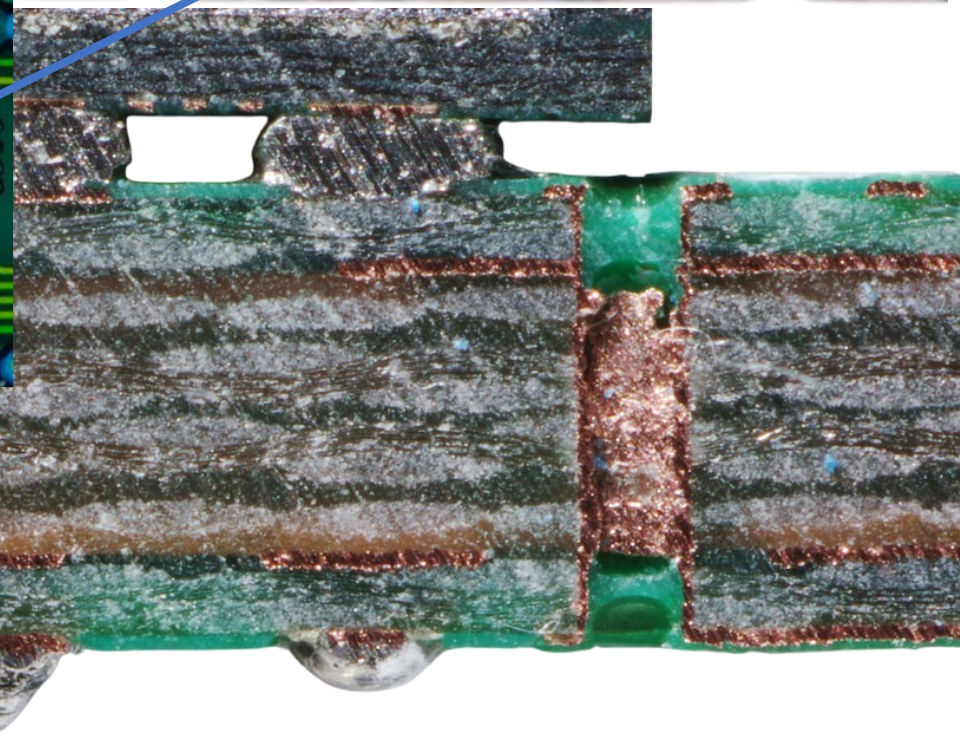




MULTICAPA

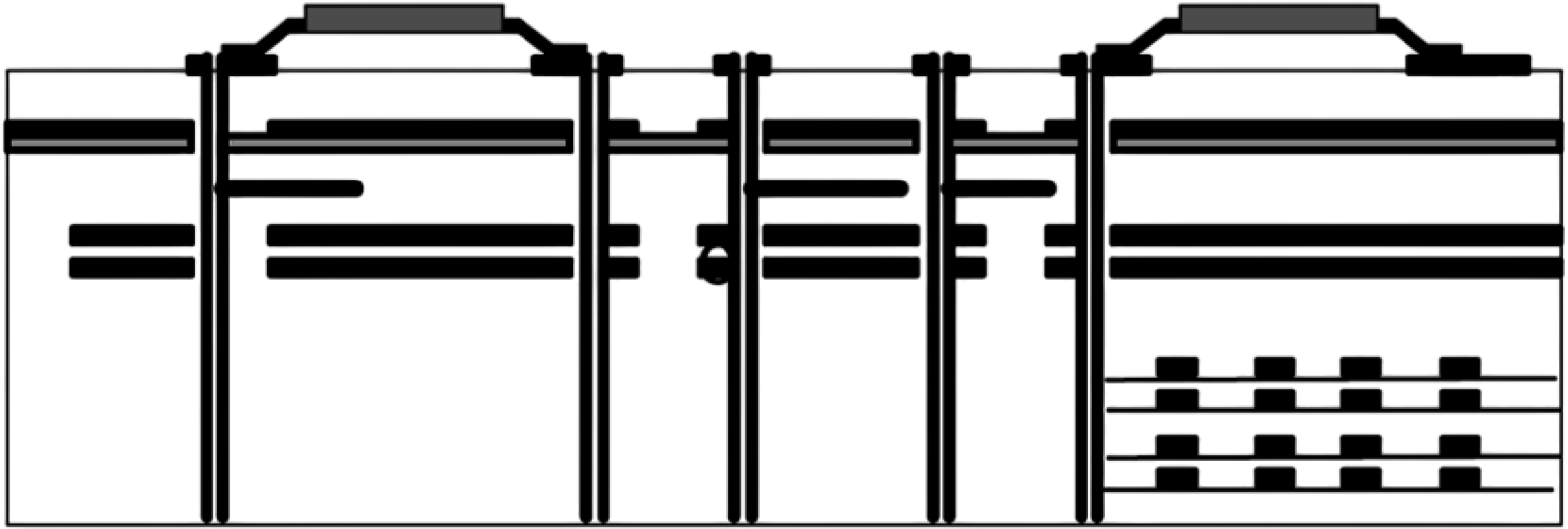




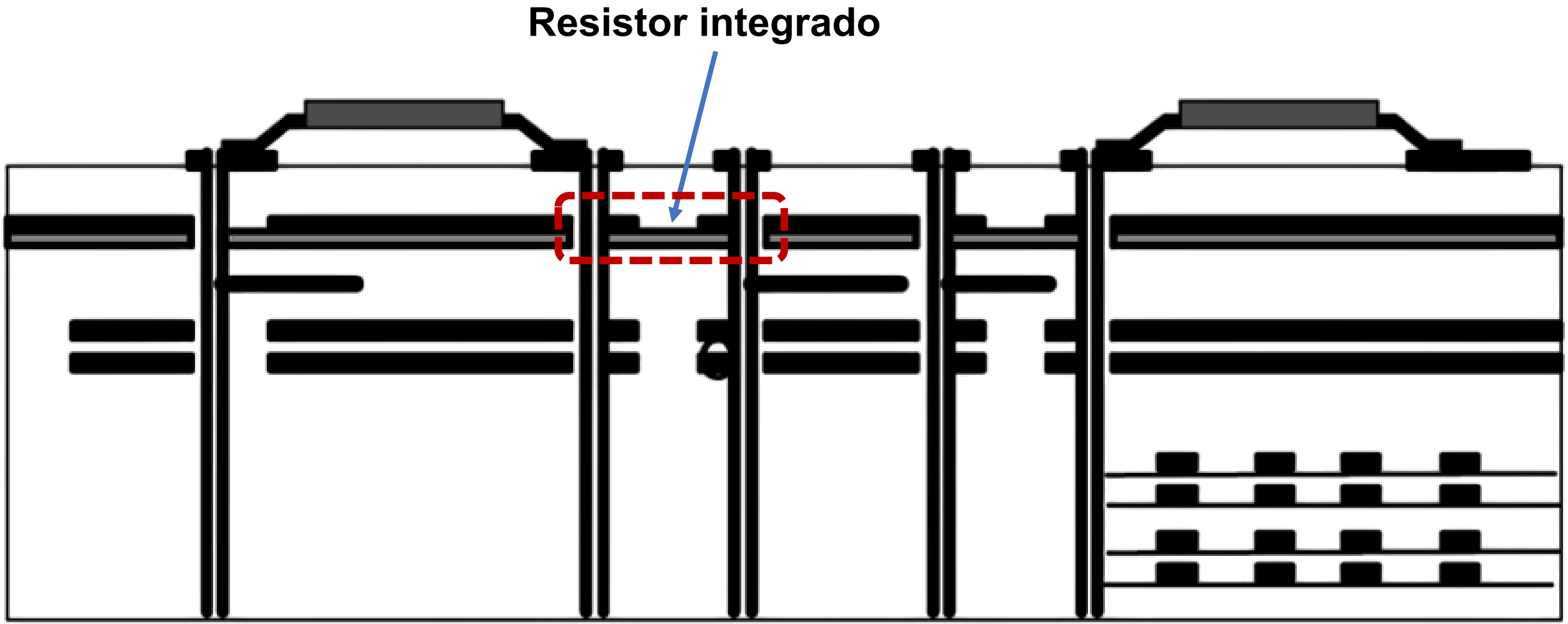


Alineación

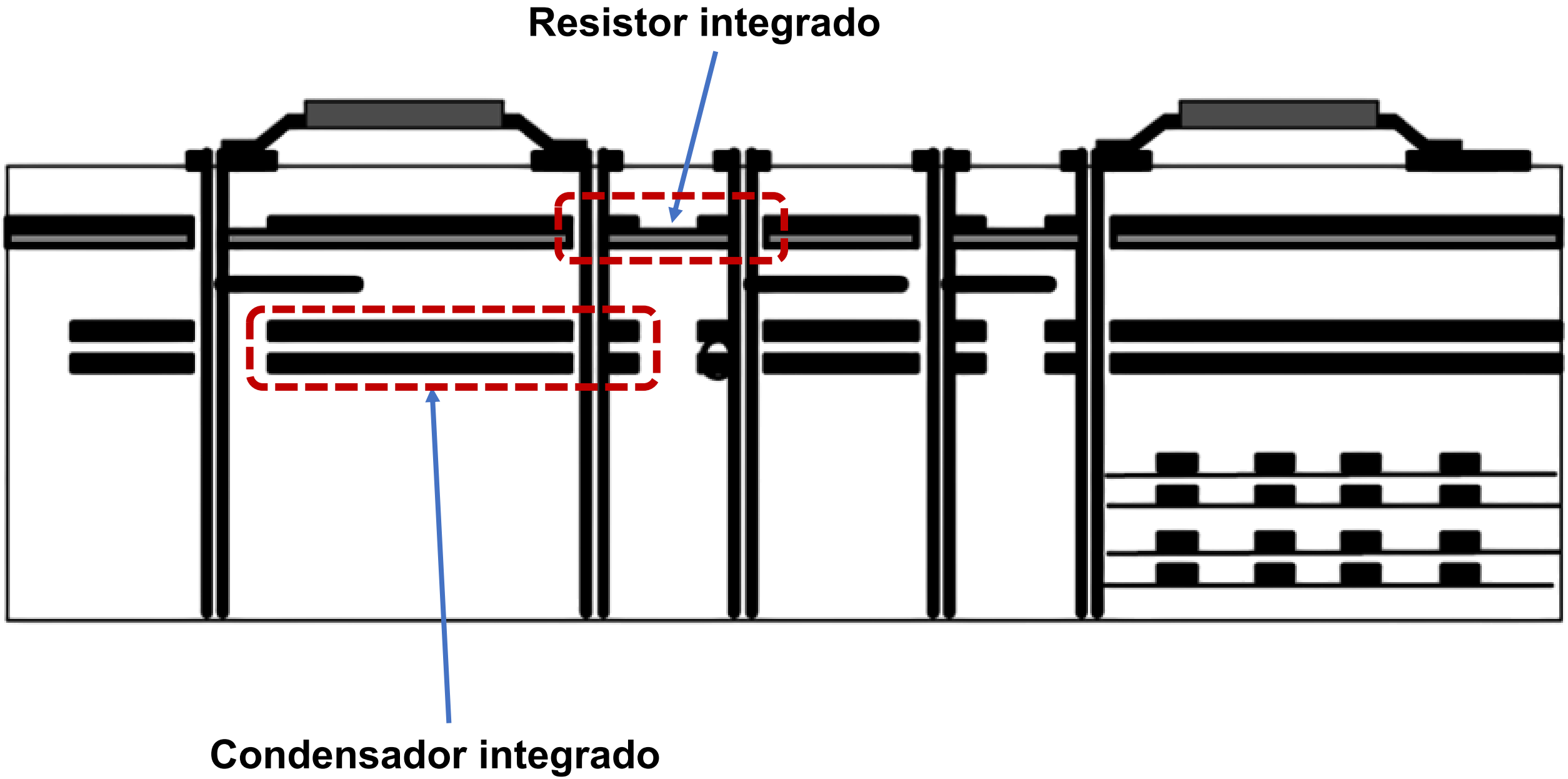
MULTICAPA – COMPONENTES PASIVOS



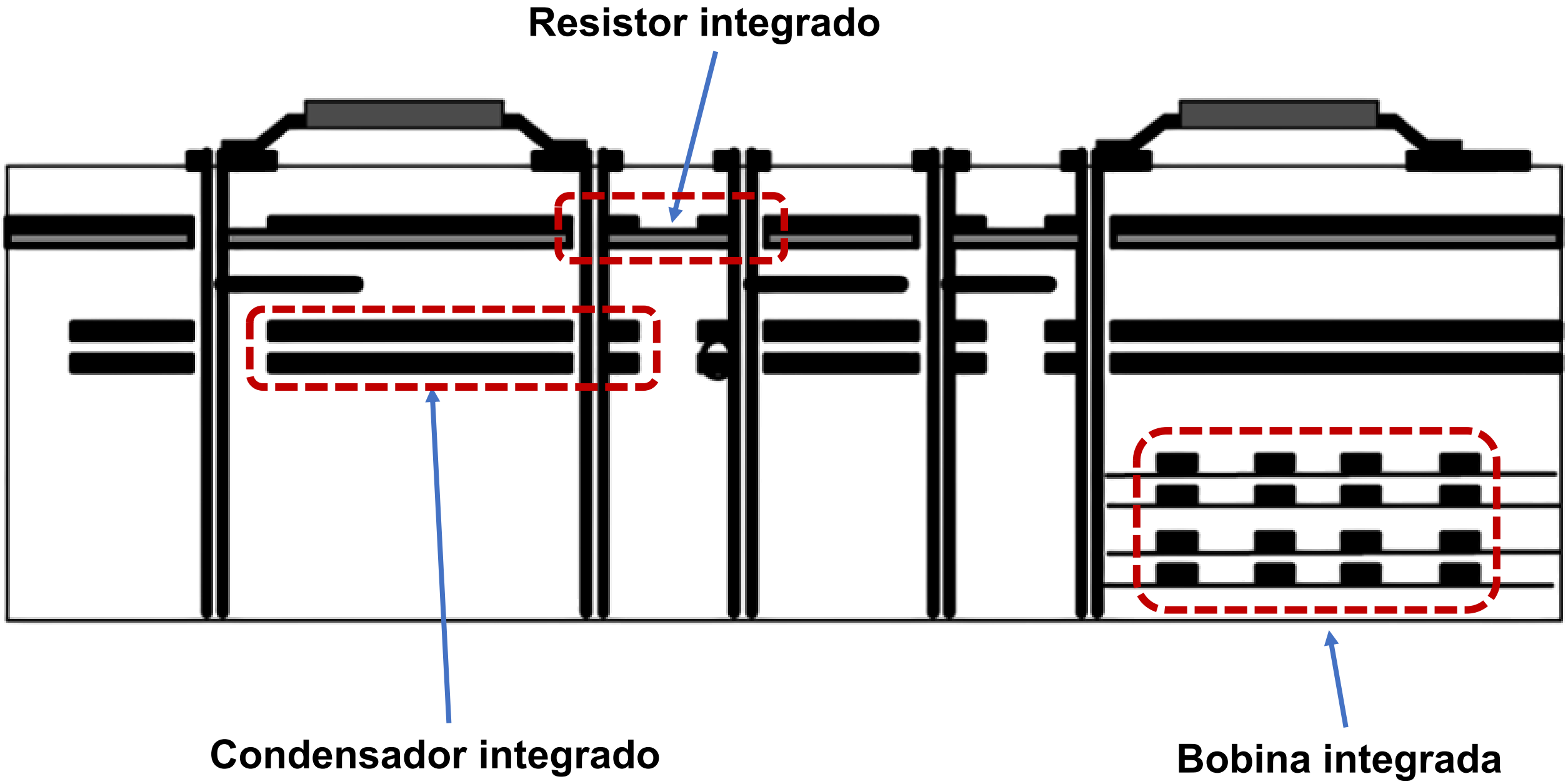
MULTICAPA – COMPONENTES PASIVOS



MULTICAPA – COMPONENTES PASIVOS

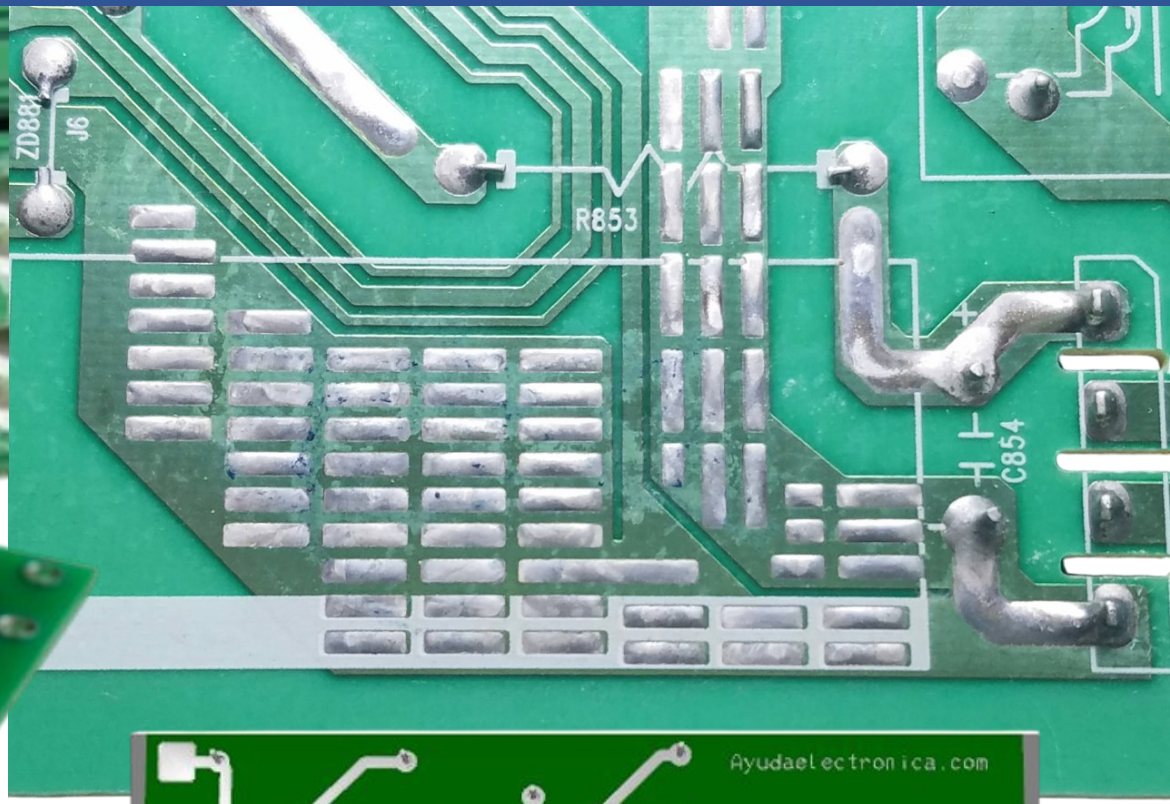


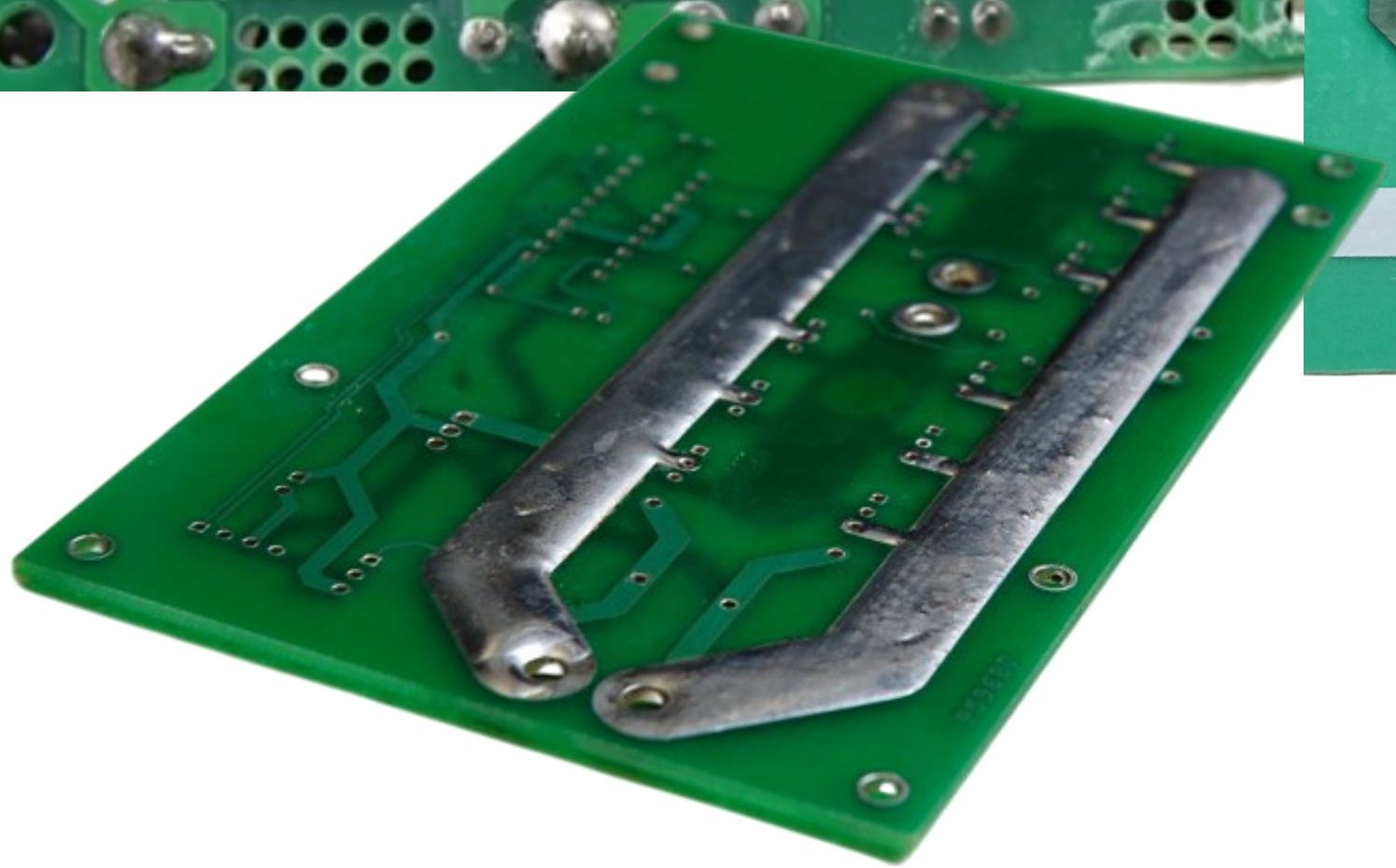
MULTICAPA – COMPONENTES PASIVOS

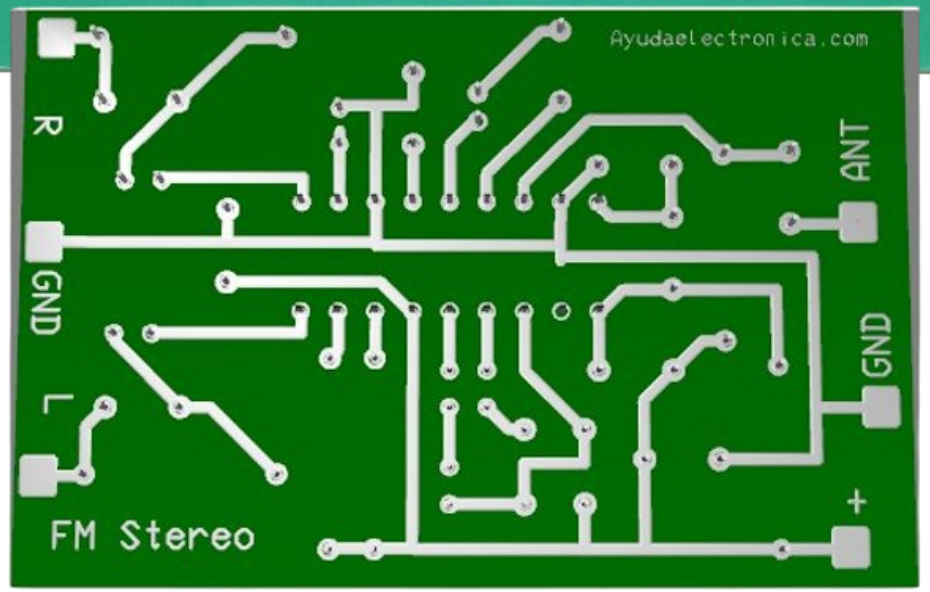


PISTAS ESTAÑADAS

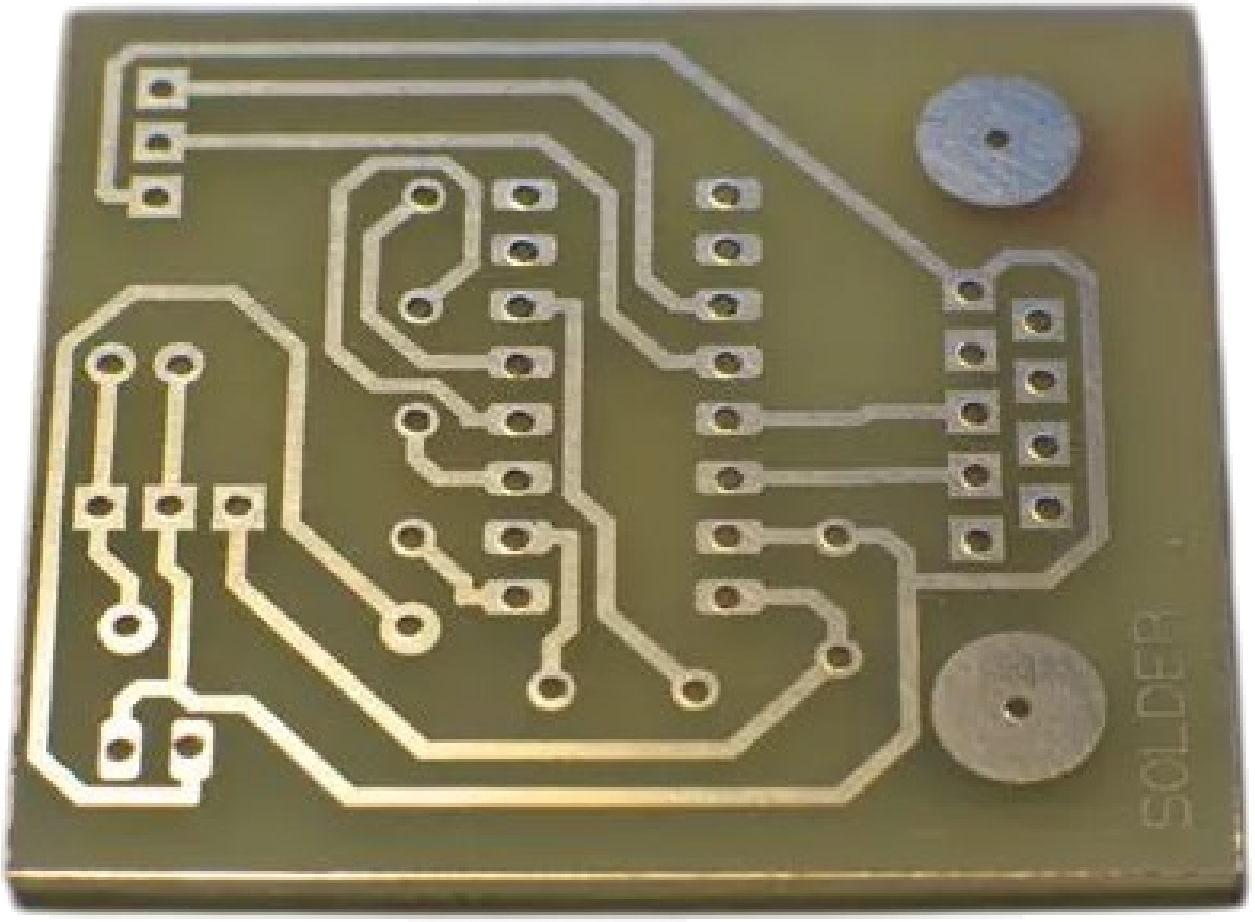


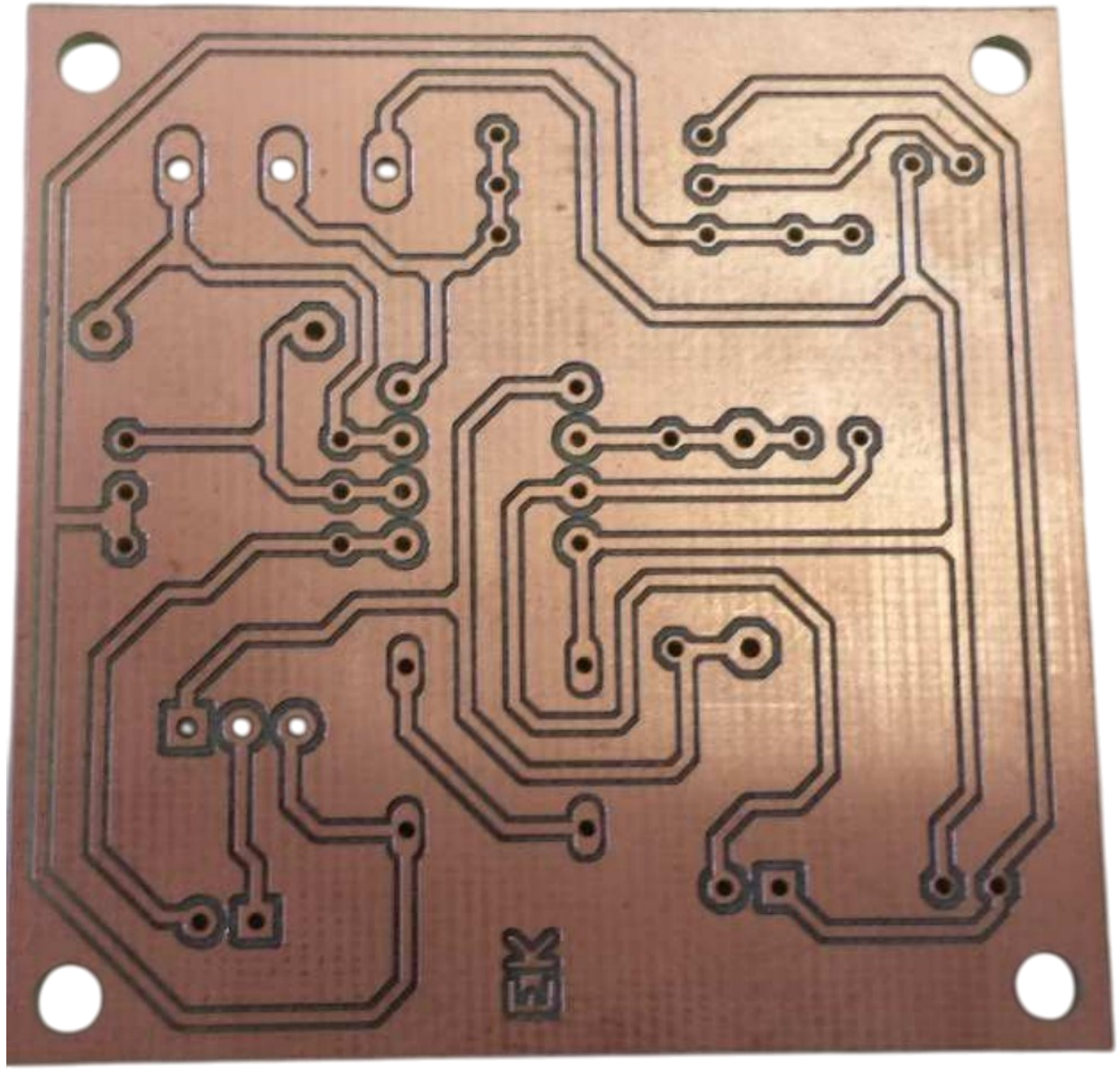






ZONAS DE COBRE

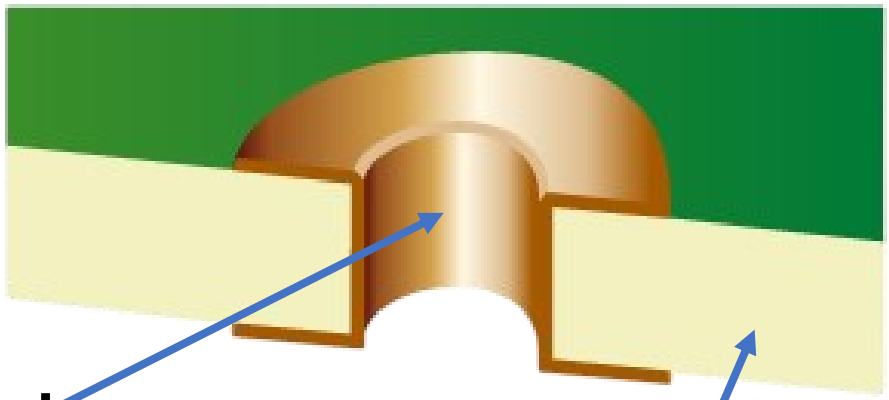


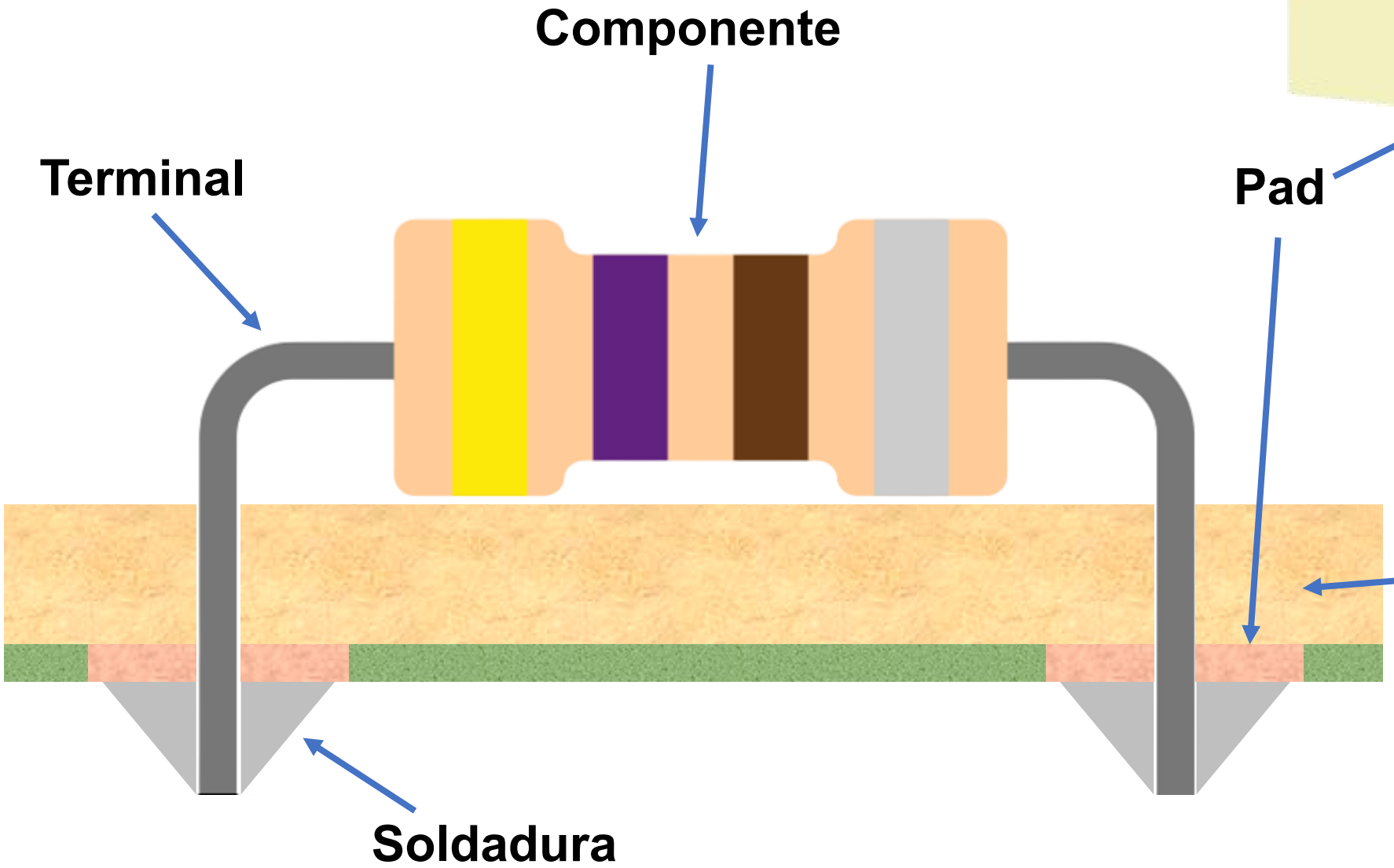


THROUGH-HOLE TECHNOLOGY

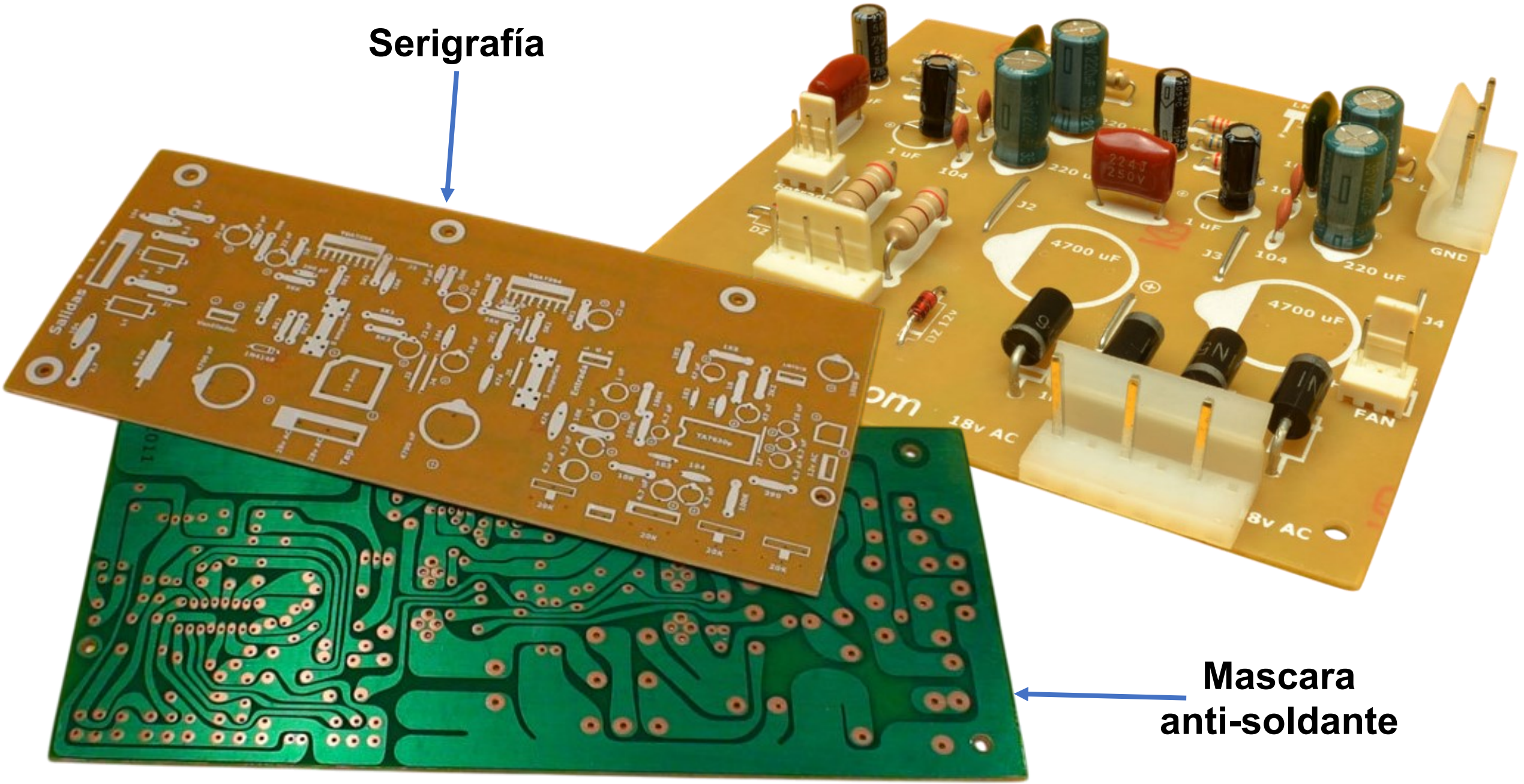


THROUGH-HOLE TECHNOLOGY

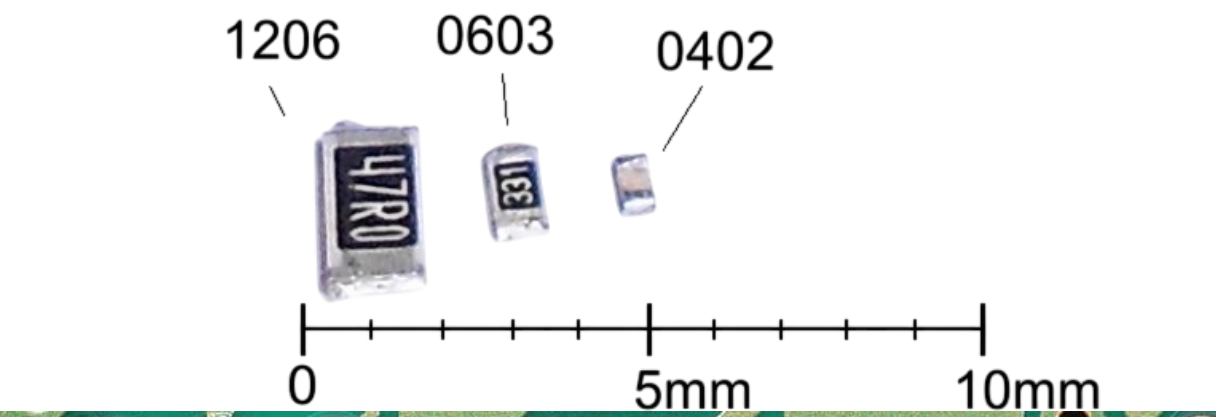


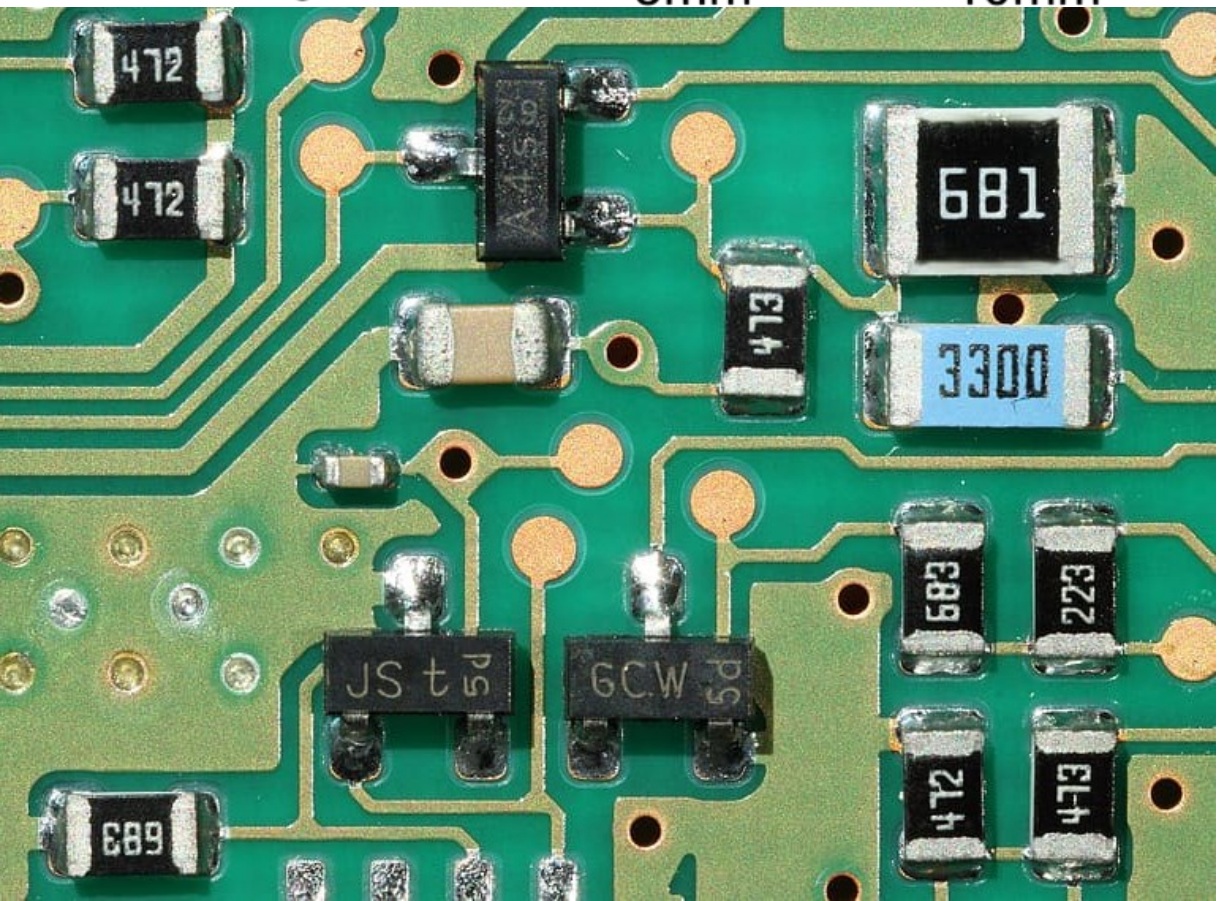


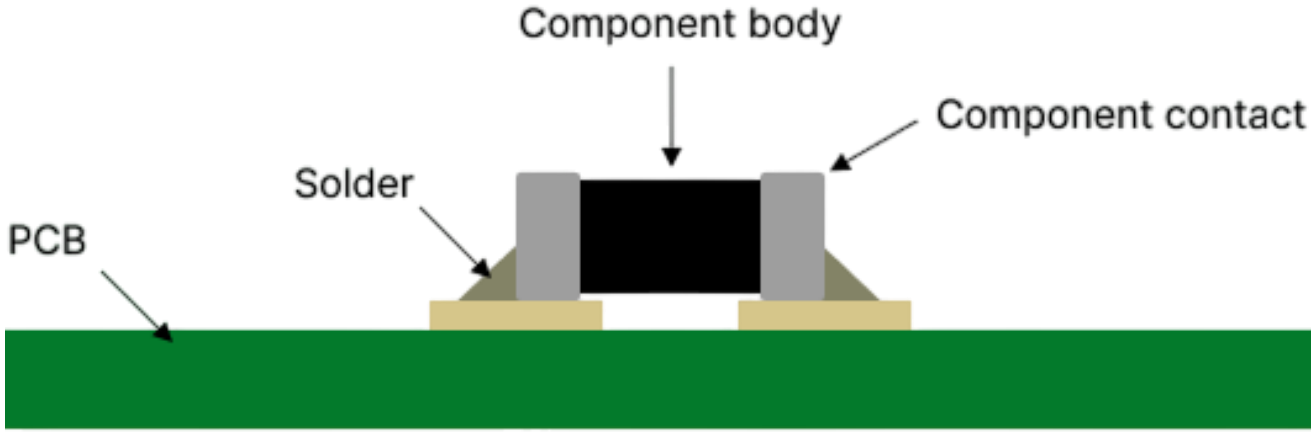
THROUGH-HOLE TECHNOLOGY

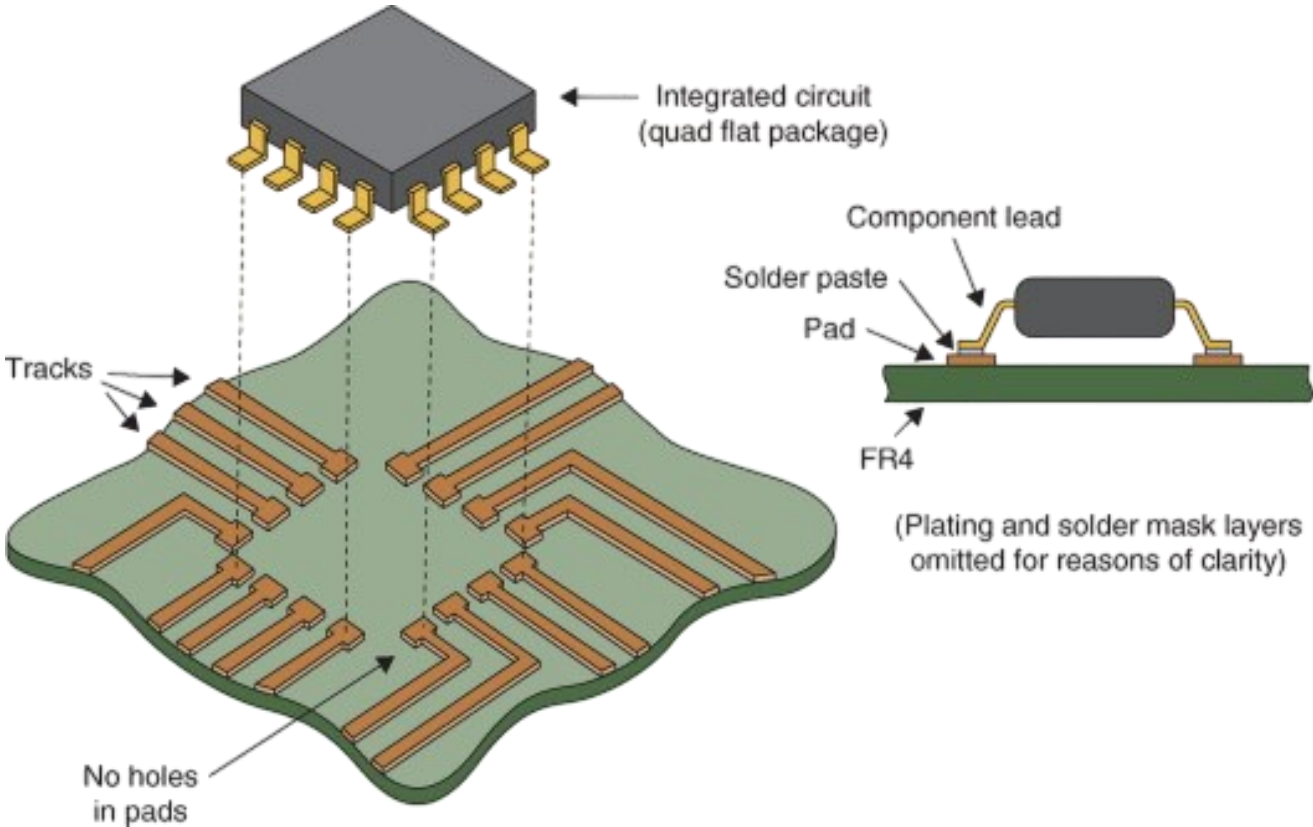


SMD TECHNOLOGY







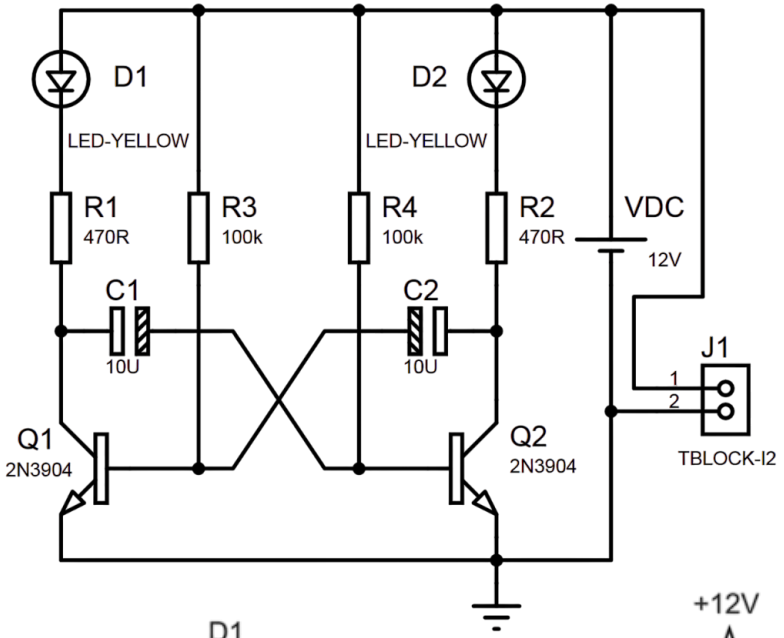


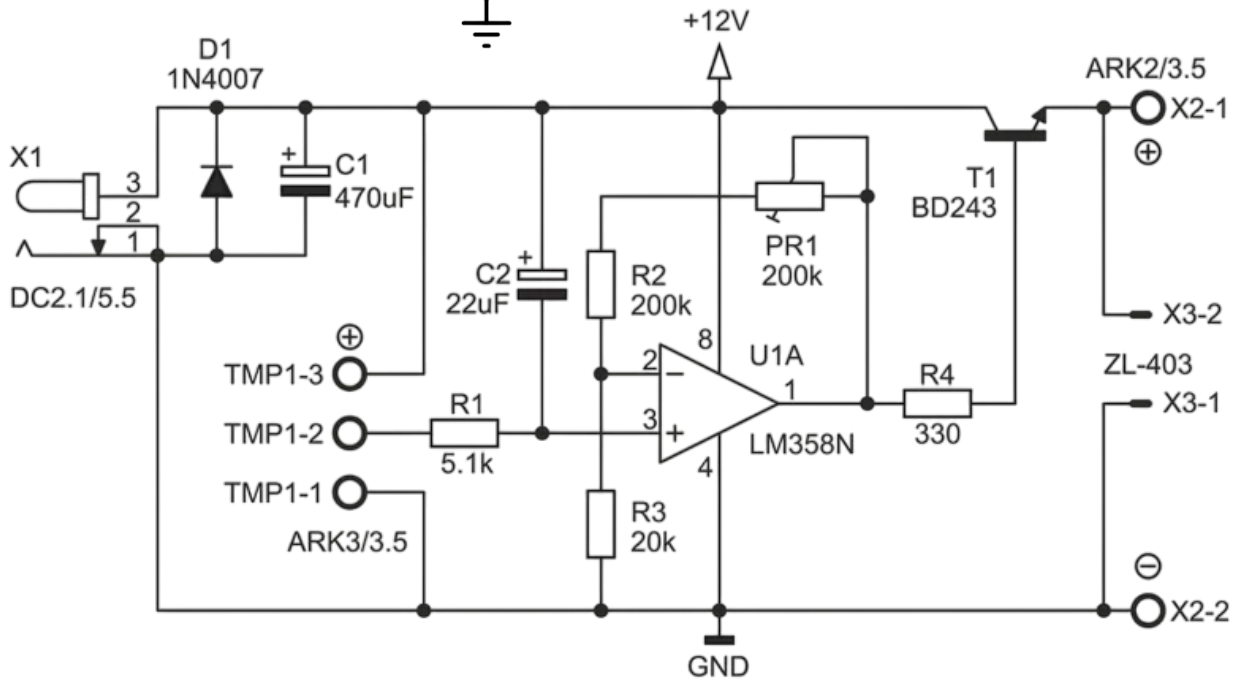
DISEÑO DEL PCB

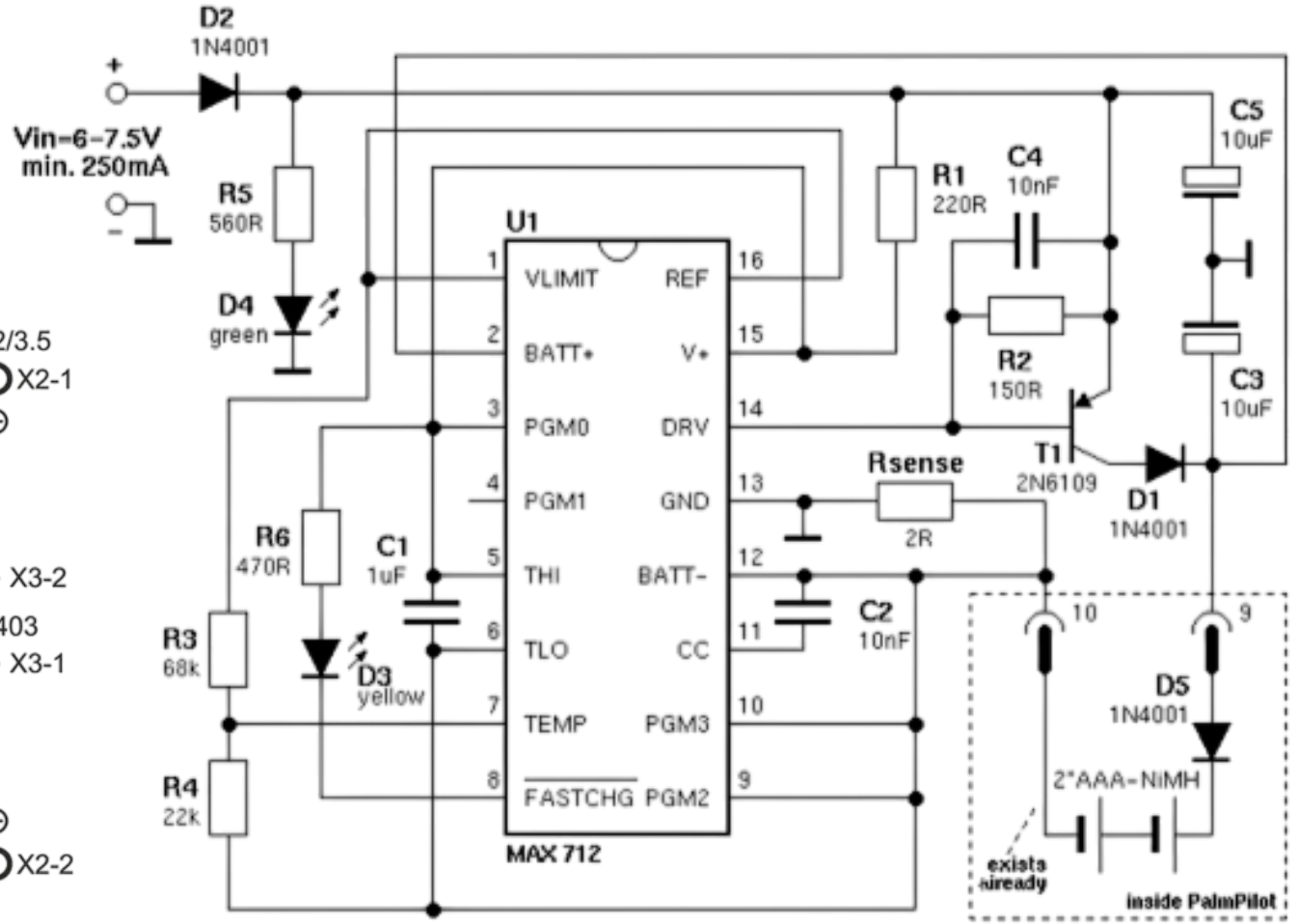
1. Creación del Esquemático (SCH)

DISEÑO DEL PCB

1. Creación del Esquemático (SCH)





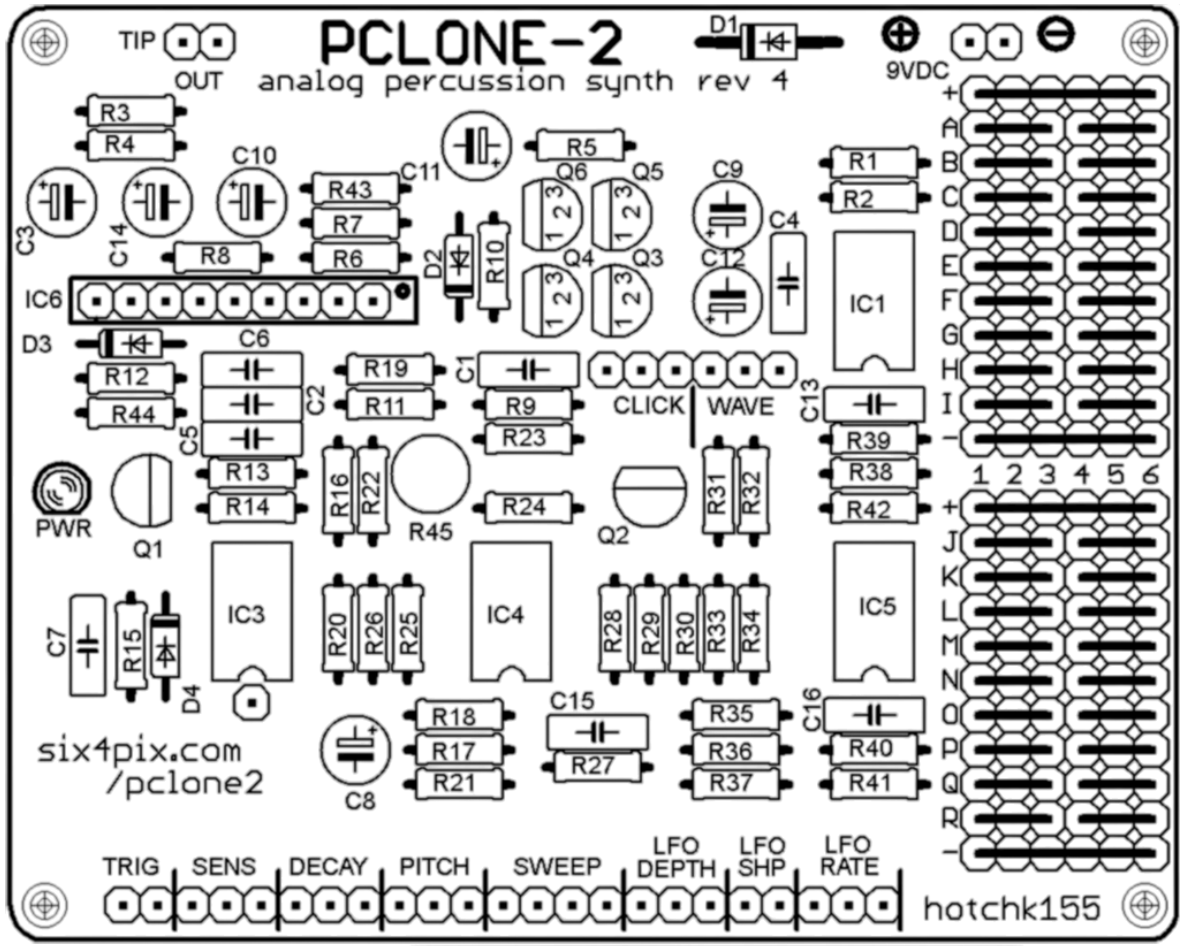


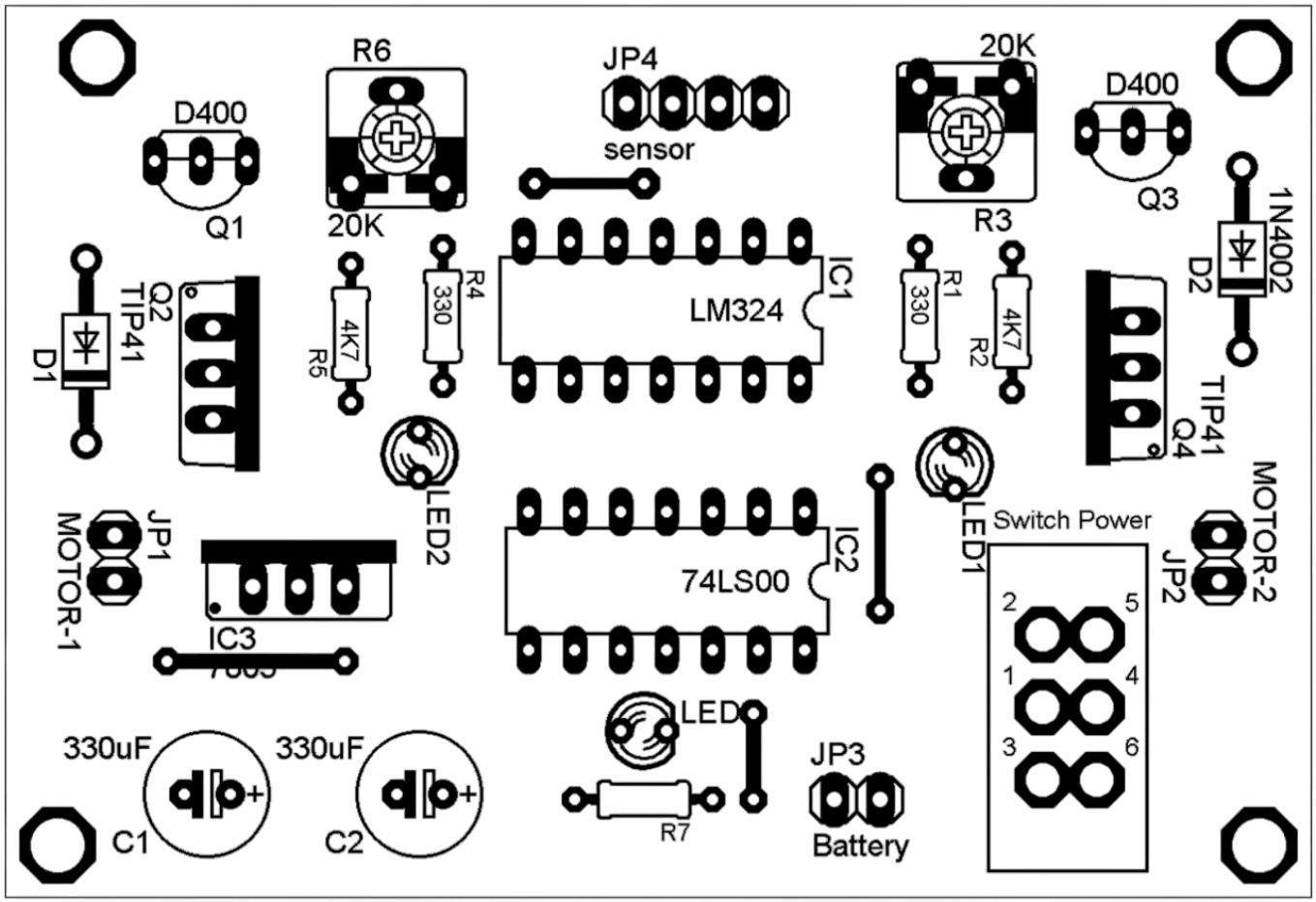
(c) 1999 Till Harbaum

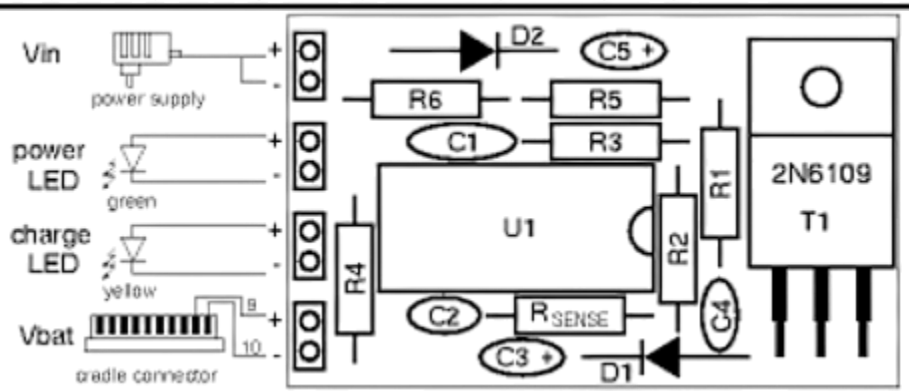
2. Ubicación de Componentes (Placement)

DISEÑO DEL PCB

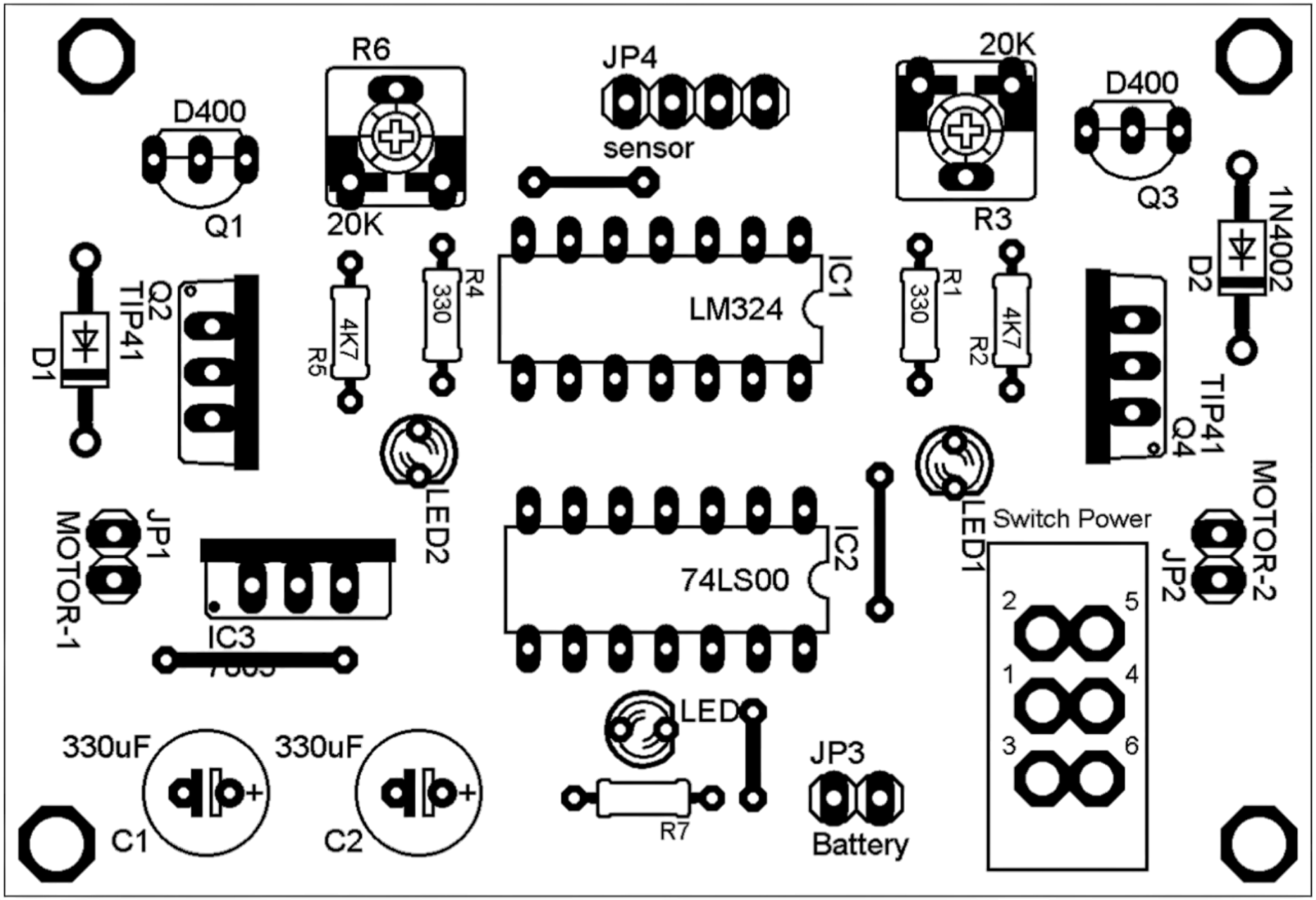
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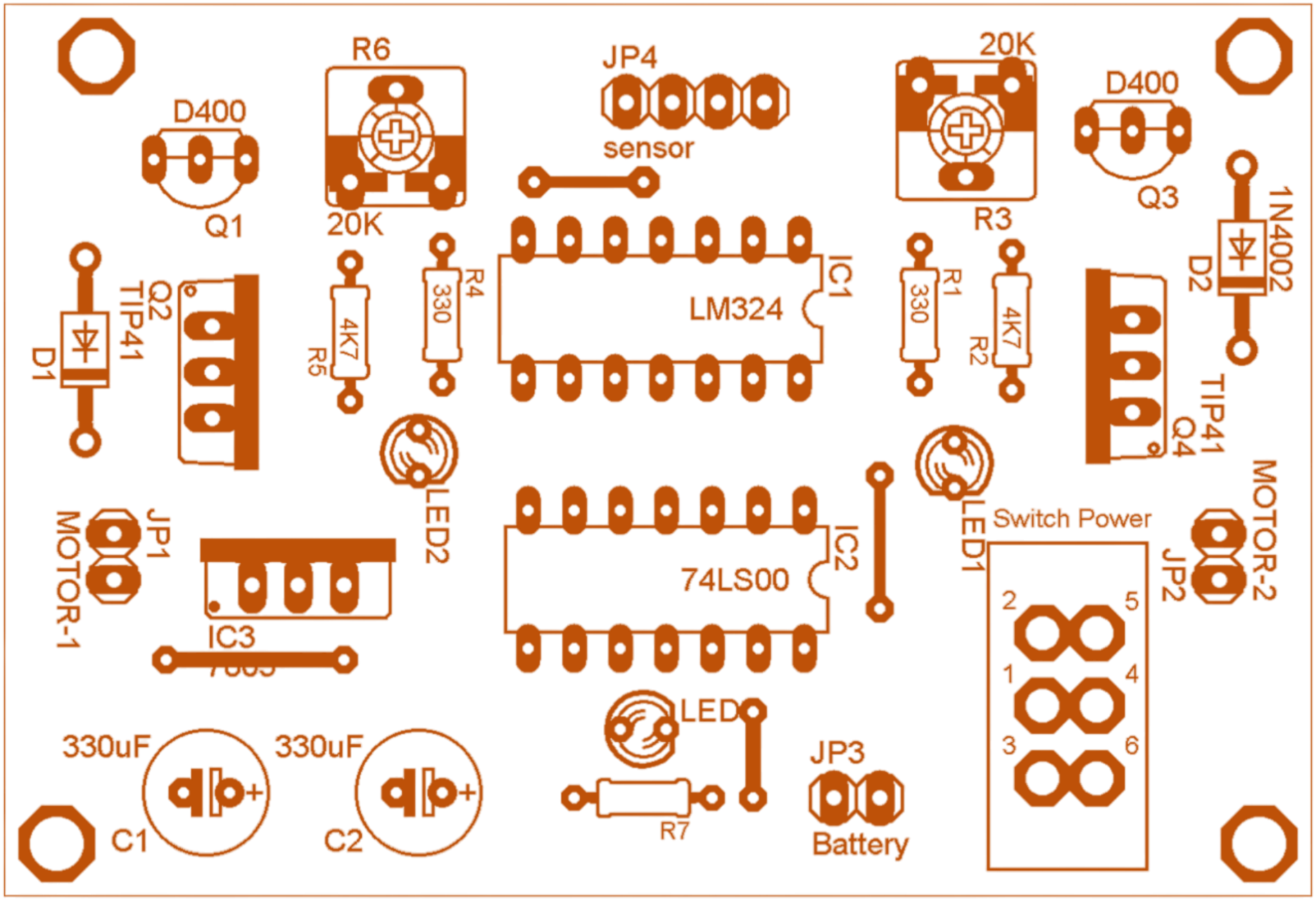




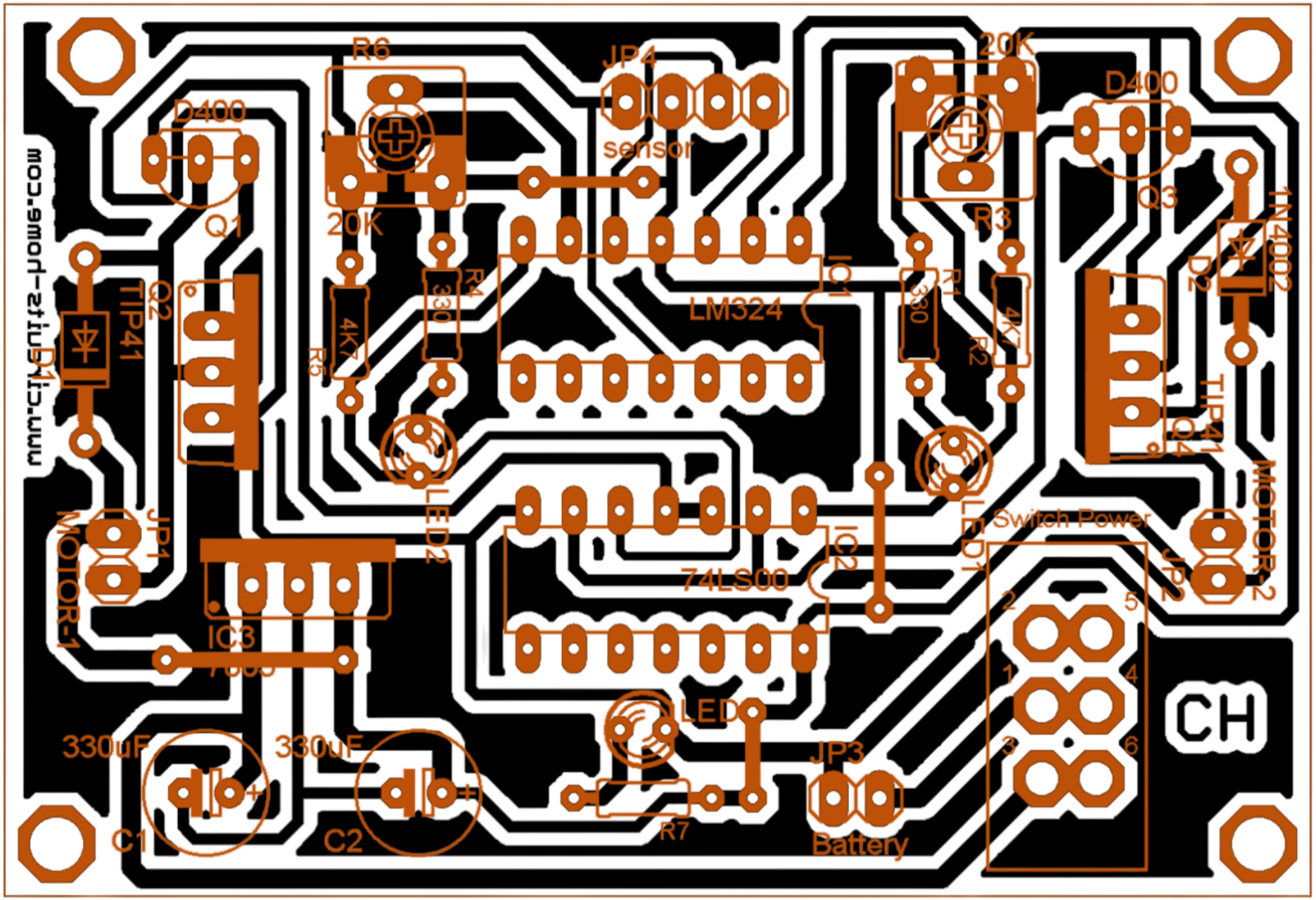
DISEÑO DEL PCB



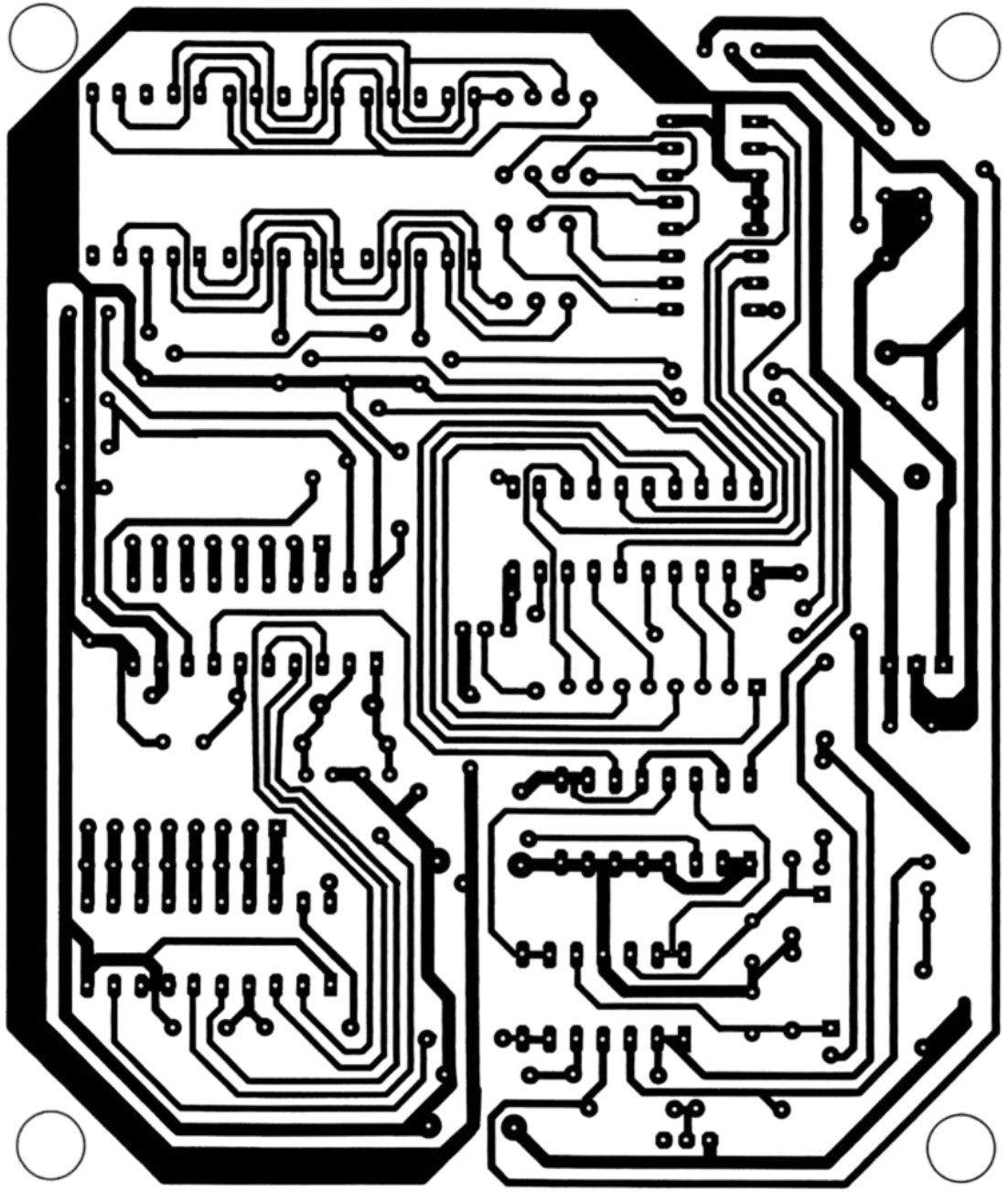
DISEÑO DEL PCB



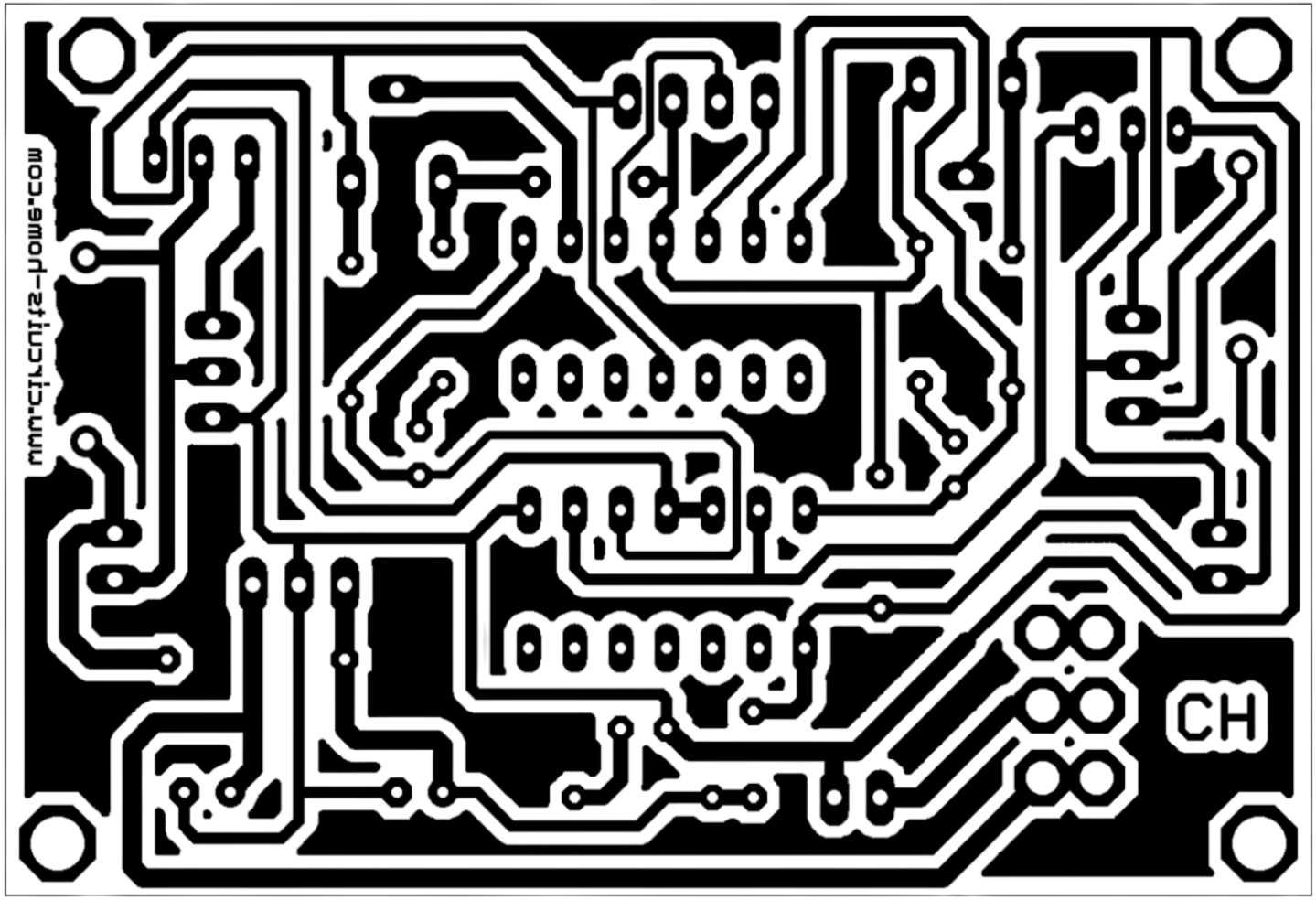
DISEÑO DEL PCB



DISEÑO DEL PCB



3. Ruteado (Routing)



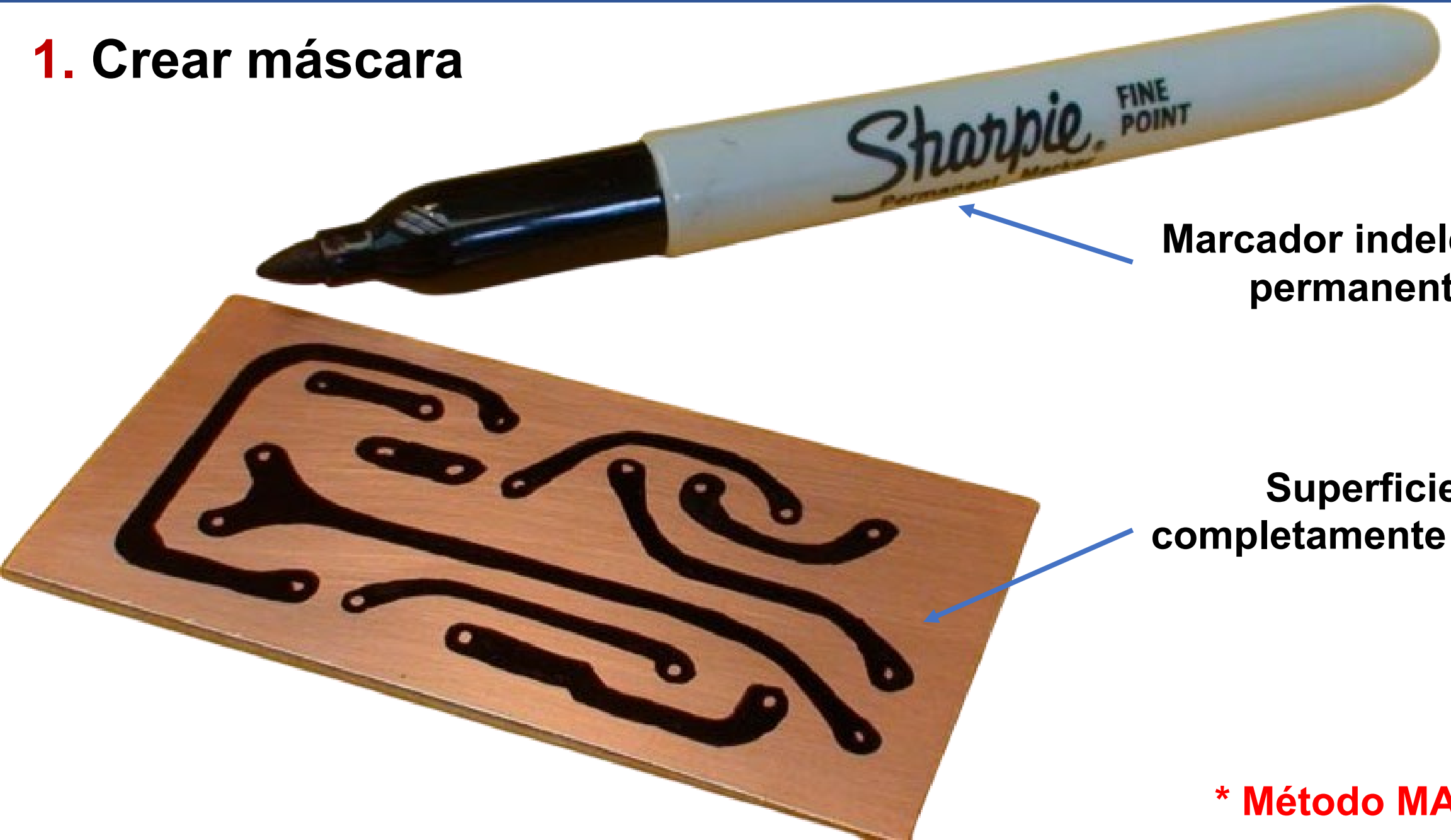
FABRICACIÓN DEL PCB

FABRICACIÓN DEL PCB

1. Crear máscara

FABRICACIÓN DEL PCB

1. Crear máscara



Marcador indeleble o permanente

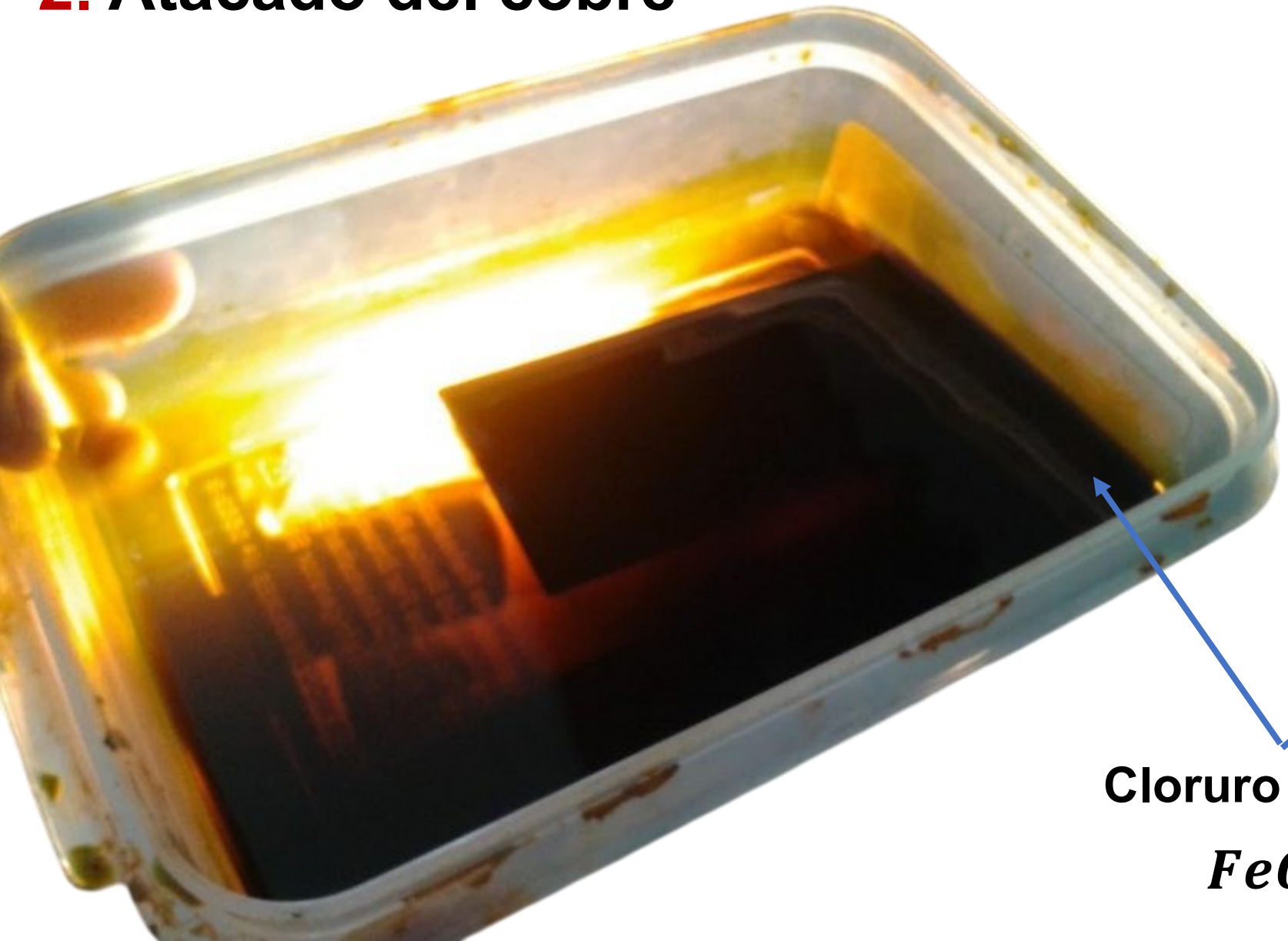
Superficie completamente limpia

* Método MANUAL

2. Atacado del cobre

FABRICACIÓN DEL PCB

2. Atacado del cobre



Cloruro Férrico

$FeCl_3$



FABRICACIÓN DEL PCB

2. Atacado del cobre



FABRICACIÓN DEL PCB

3. Limpieza

FABRICACIÓN DEL PCB

3. Limpieza



Jabón blanco
detiene la acción corrosiva

FABRICACIÓN DEL PCB

3. Limpieza



Jabón blanco
detiene la acción corrosiva



Rollitos de viruta
para quitar el marcador

FABRICACIÓN DEL PCB

3. Limpieza

Jabón blanco
detiene la acción corrosiva



Rollitos de viruta
para quitar el marcador



Alcohol isopropílico/etílico
quita impurezas y grasitudes

FABRICACIÓN DEL PCB

3. Limpieza

Jabón blanco
detiene la acción corrosiva



Rollitos de viruta
para quitar el marcador





Alcohol isopropílico/etílico
quita impurezas y grasitudes

FABRICACIÓN DEL PCB

4. Perforado

FABRICACIÓN DEL PCB

4. Perforado



FABRICACIÓN DEL PCB

4. Perforado



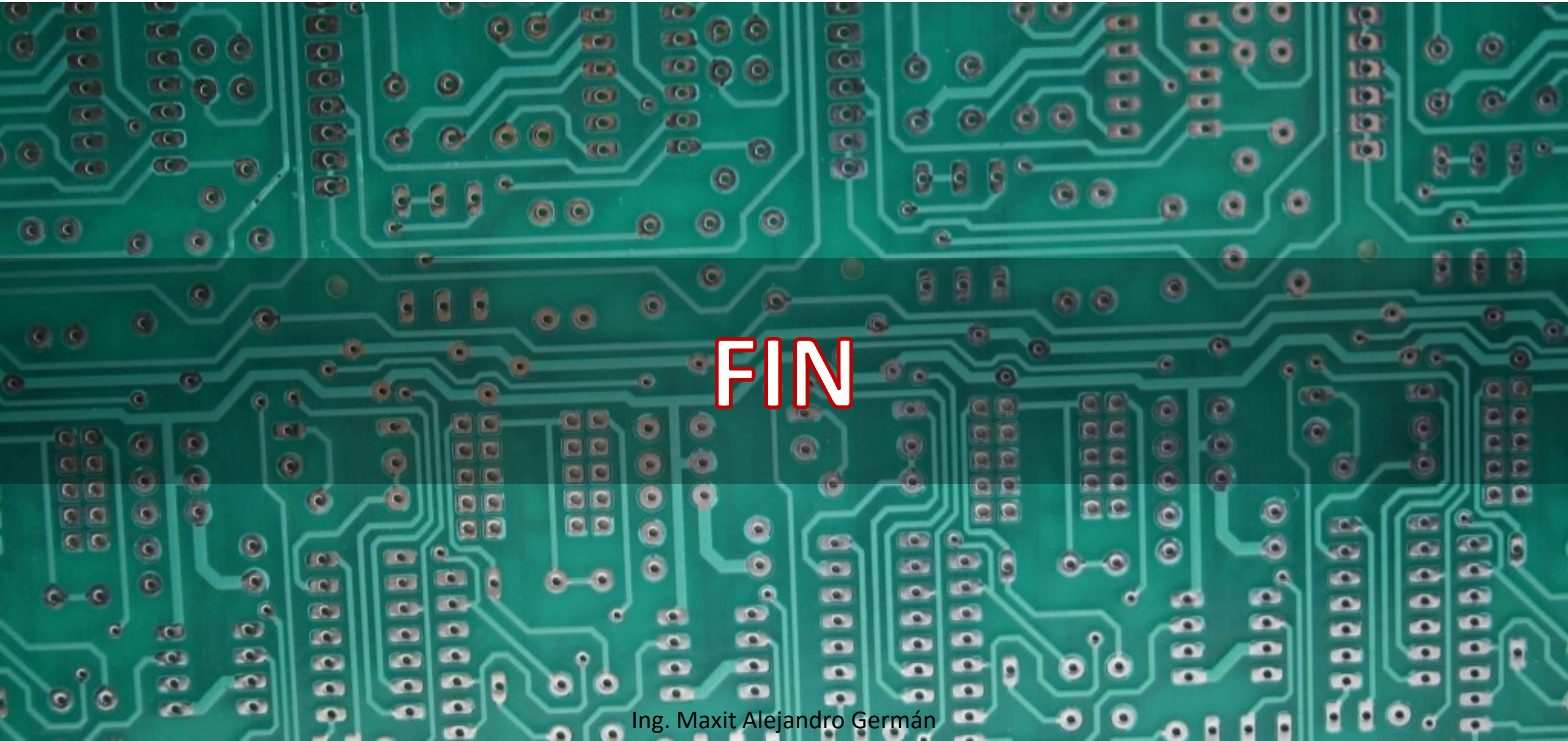




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FIN

Ing. Maxit Alejandro Germán